

# Bayesian Hierarchical Poisson Model

## Load the data

```
library("haven")
library("dplyr")

##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##     filter, lag
##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

wm <- read_sav("wm.sav")
wm <- wm %>% arrange(HH1, HH2, WM3)
hh <- read_sav("hh.sav")
df_hh <- hh %>% select(HH1, HH2, HHSEX, HHAGE)
wm <- merge(x=wm, y=df_hh, by = c("HH1", "HH2"))
```

## Clean the data

(one line of code)

```
wm <- wm %>% filter(MA1 == 1 & MT1!=9 & MT2!=9 & MT3!=9 & MA2!=98 & MA2!=99 & MA3!=9)
```

68709 women before cleaning the data

MA1: "Currently married"

```
table(wm$MA1,useNA="ifany")
```

```
##
##      1
## 51331
```

```
wm <- wm %>% filter(MA1==1)
```

```
1      3  <NA>
51426 12951 4332
```

We exclude 12951 women who have never married and 4332 women with unknown status. In total, 17283 samples were excluded.

Technology usage (22): Missing data: media\_newspaper (5), media\_radio (3), media\_tv(14)

```
table(wm$MT1,useNA="ifany")
```

```
##
##      0      1      2      3
## 48411 1364   884   672
```

```
table(wm$MT2,useNA="ifany")
```

```
##
##      0      1      2      3
## 50608   316   271   136
```

```
table(wm$MT3,useNA="ifany")
```

```
##
##      0      1      2      3
## 18295 2321  4162 26553
```

```
attr(wm$MT1, "labels")
```

```
##          NOT AT ALL LESS THAN ONCE A WEEK  AT LEAST ONCE A WEEK
##                                0                1                2
##      ALMOST EVERY DAY          NO RESPONSE
##                                3                9
```

```
wm <- wm %>% filter(MT1!=9 & MT2!=9 & MT3!=9)
```

other wives: (30)

```
table(wm$MA3,useNA="ifany")
```

```
##
##      1      2
## 1618 49713
```

```
wm <- wm %>% filter(MA3!=9)
```

Age of husband (44) (98 for "DK"; 99 for "NO RESPONSE")

```
table(wm$MA2,useNA="ifany")
```

```
##
##  15  16  17  18  19  20  21  22  23  24  25  26  27  28  29  30
##   1   3  14  73  95 237 302 479 527 742 1038 1042 1055 1448 1201 2178
##  31  32  33  34  35  36  37  38  39  40  41  42  43  44  45  46
## 1495 1909 1456 1556 2558 1828 1567 2035 1588 2212 1485 1559 1162 1148 1822 1375
##  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61  62
## 1368 1620 1137 1429 1029 978 819 715 1051 801 553 532 343 544 246 199
##  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78
## 150 105 164 72  41  51  19  59  22  16   8   9  21  10   5   6
##  79  80  82  84  85  88  90
##   7   2   3   1   2   1   3
```

```
wm <- wm %>% filter(MA2!=98 & MA2!=99)
```

HHAGE

```
table(wm$HHAGE,useNA="ifany")
```

```
##
##  16  17  18  19  20  21  22  23  24  25  26  27  28  29  30  31
##   3   7  28  40  89 154 231 258 383 590 624 674 934 846 1470 1142
##  32  33  34  35  36  37  38  39  40  41  42  43  44  45  46  47
```

```
## 1327 1158 1311 1863 1588 1427 1732 1491 1799 1445 1399 1146 1113 1640 1376 1435
## 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
## 1633 1212 1450 1198 1078 984 902 1320 1108 810 835 705 1095 611 606 511
## 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79
## 363 649 379 308 364 218 499 257 212 145 94 211 128 95 102 61
## 80 81 82 83 84 85 86 87 88 89 90 92 93 94 95
## 137 55 42 33 21 56 26 27 10 9 19 9 1 2 18
```

No missing values for the following predictors:

```
table(wm$HH6,useNA="ifany")
```

```
##
## 1 2
## 10123 41208
```

```
table(wm$HH7,useNA="ifany")
```

```
##
## 10 20 30 40 45 50 55 60
## 4539 9135 10364 8410 2670 6363 6430 3420
```

```
table(wm$HH7A,useNA="ifany")
```

```
##
## 1 3 4 6 9 10 12 13 15 18 19 22 26 27 29 30
## 804 600 770 726 749 790 765 677 1675 876 1094 686 1963 804 720 816
## 32 33 35 36 38 39 41 42 44 46 47 48 49 50 51 52
## 707 644 642 878 779 588 882 707 861 642 755 629 808 899 695 819
## 54 55 56 57 58 59 61 64 65 67 68 69 70 72 73 75
## 618 817 696 885 874 782 844 769 775 811 756 754 835 657 846 839
## 76 77 78 79 81 82 84 85 86 87 88 89 90 91 93 94
## 816 806 847 740 802 746 646 739 657 856 818 581 831 837 700 901
```

```
#table(wm$wmweight,useNA="ifany")
```

```
table(wm$WAGEM,useNA="ifany")
```

```
##
## 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
## 1 2 1 5 16 40 614 1730 3813 5751 7125 7463 6783 5599 3721 2647
## 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36
## 1764 1252 889 559 473 332 221 169 105 77 56 29 35 15 8 9
## 37 38 39 41 42 43 44 45 48
## 8 3 4 4 3 1 2 1 1
```

```
table(wm$WB4,useNA="ifany")
```

```
##
## 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
## 183 356 450 1359 1483 1580 1509 1669 1707 1745 1838 1955 1753 1858 1762 2205
## 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46
## 2015 1900 1848 1803 1799 2045 1630 1777 1480 1433 1405 1293 1172 1084 1021 1190
## 47 48 49
## 1135 1153 736
```

```
table(wm$windex5,useNA="ifany")
```

```
##
## 1 2 3 4 5
```

```
## 10718 10732 10702 10336 8843
```

```
table(wm$welevel,useNA="ifany")
```

```
##
```

```
##      0      1      2      3
```

```
## 9186 13235 22289 6621
```

```
table(wm$ethnicity,useNA="ifany")
```

```
##
```

```
##      1      2
```

```
## 50184 1147
```

```
table(wm$CM11,useNA="ifany")
```

```
##
```

```
##      0      1      2      3      4      5      6      7      8      9     10     11     12
```

```
## 4663 10933 16565 10670 5029 2029 885 337 145 52 18 3 2
```

## Summary

```
women_age <- wm$WB4
women_age2 <- women_age^2
husband_age <- wm$MA2
#husband_age2 <- husband_age^2
HH_age <- wm$HHAGE
#HH_age2 <- wm$HHAGE^2
women_agem <- wm$WAGEM
division <- wm$HH7
ethnicity <- wm$ethnicity # Fixed
area <- wm$HH6 # Fixed
women_edu <- wm$welevel
HH_sex <- wm$HHSEX
other_wives <- wm$MA3
windex <- wm$windex5
media_newspaper <- ifelse(wm$MT1>0, 1, 0)
media_radio <- ifelse(wm$MT2>0, 1, 0)
media_tv <- ifelse(wm$MT3>0, 1, 0)
media <- media_newspaper|media_radio|media_tv
media <- ifelse(media,1,0)
# Response variable
CEB <- wm$CM11
# District
district <- wm$HH7A
# Sampling weight
wmweight <- wm$wmweight

mean(women_age)
```

```
## [1] 32.02186
```

```
sd(women_age)
```

```
## [1] 8.701892
```

```

mean(husband_age)

## [1] 39.94711
sd(husband_age)

## [1] 10.29503
mean(HH_age)

## [1] 45.12719
sd(HH_age)

## [1] 12.78071
mean(women_agem)

## [1] 16.85521
sd(women_agem)

## [1] 3.300367
total <- dim(wm)[1]
table(division)

## division
##      10      20      30      40      45      50      55      60
## 4539  9135 10364  8410  2670  6363  6430  3420
round(table(division)/total,4)

## division
##      10      20      30      40      45      50      55      60
## 0.0884 0.1780 0.2019 0.1638 0.0520 0.1240 0.1253 0.0666
table(area)

## area
##      1      2
## 10123 41208
round(table(area)/total,4)

## area
##      1      2
## 0.1972 0.8028
attr(wm$welevel, "labels")

## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9
table(women_edu)

## women_edu
##      0      1      2      3
## 9186 13235 22289 6621

```

```
round(table(women_edu)/total,4)
```

```
## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
```

```
attr(wm$welevel, "labels")
```

```
## Pre-primary or none      Primary      Secondary      Higher secondary+
##              0              1              2              3
##      Missing/DK
##              9
```

```
table(women_edu)
```

```
## women_edu
##      0      1      2      3
## 9186 13235 22289 6621
```

```
round(table(women_edu)/total,4)
```

```
## women_edu
##      0      1      2      3
## 0.1790 0.2578 0.4342 0.1290
```

```
attr(wm$HHSEX, "labels")
```

```
##      Male  Female Missing
##          1      2      9
```

```
table(wm$HHSEX)
```

```
##
##      1      2
## 47345 3986
```

```
round(table(wm$HHSEX)/total,4)
```

```
##
##      1      2
## 0.9223 0.0777
```

```
attr(other_wives, "labels")
```

```
##      YES      NO NO RESPONSE
##          1          2          9
```

```
table(other_wives)
```

```
## other_wives
##      1      2
## 1618 49713
```

```
round(table(other_wives)/total,4)
```

```
## other_wives
##      1      2
## 0.0315 0.9685
```

```
attr(windex, "labels")
```

```
## Poorest  Second  Middle  Fourth Richest
```

```

##      1      2      3      4      5
table(windex)

## windex
##      1      2      3      4      5
## 10718 10732 10702 10336  8843

round(table(windex)/total,4)

## windex
##      1      2      3      4      5
## 0.2088 0.2091 0.2085 0.2014 0.1723

table(media)

## media
##      0      1
## 17834 33497

round(table(media)/total,4)

## media
##      0      1
## 0.3474 0.6526

attr(wm$ethnicity, "labels")

##      Bengali      Other Missing/DK
##           1           2           9

table(ethnicity)

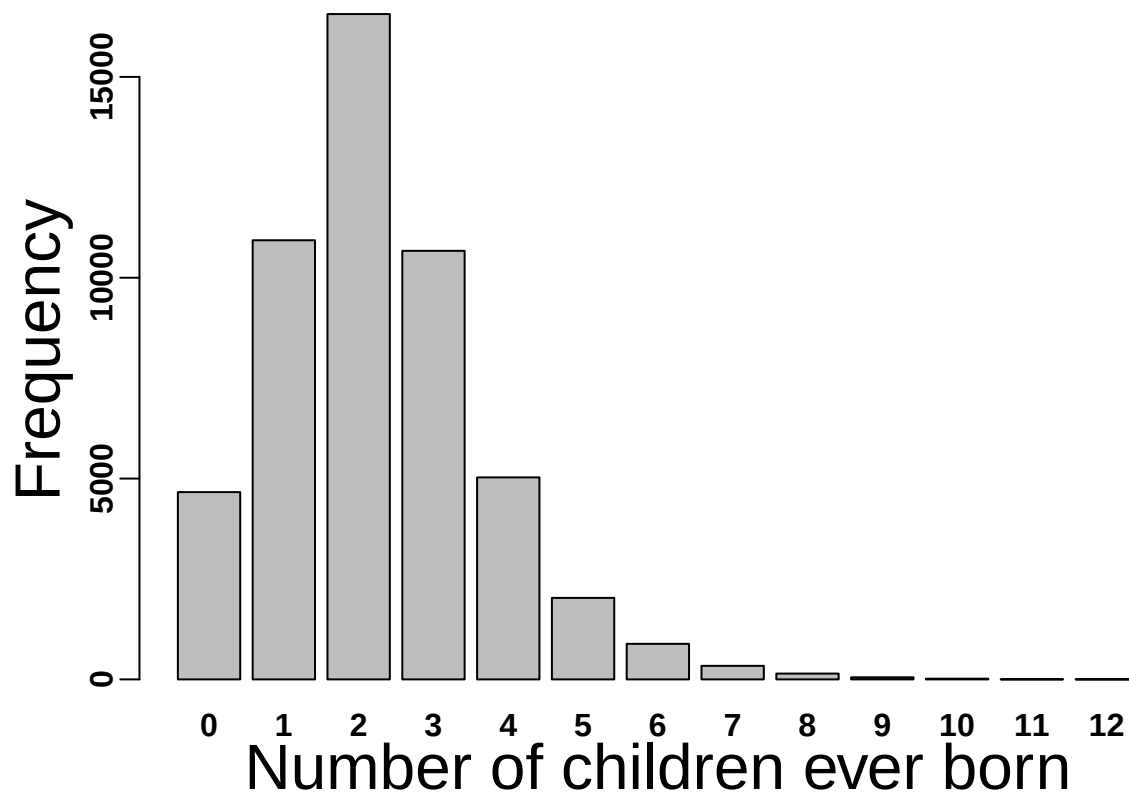
## ethnicity
##      1      2
## 50184 1147

round(table(ethnicity)/total,4)

## ethnicity
##      1      2
## 0.9777 0.0223

IMAGEOUT <- "./pics/BHP/"
par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
barplot(table(CEB), ylab="Frequency",
        xlab="Number of children ever born", cex.lab = 2, names.arg=c(0:12), font.axis = 2.5)

```



```
dev.print(pdf,  
    paste(IMAGEOUT,"CEB_histo.pdf",sep=""),  
    width=16,height=12)
```

```
## pdf
## 2
```

```
y <- CEB  
division <- factor(distribution, levels = c(10,20,30,40,45,50,55,60), labels = c("Barishal", "Chattogram",  
division <- relevel(distribution, ref=3)
```

```
division <- model.matrix(~division, data = division)
colnames(division) <- c(
  "Dhaka.ref",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymensingh",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)
```

```
district_names <- c("Bagerhat", "Bandarban", "Barguna", "Barishal", "Bhola", "Bogura", "Brahmanbaria",
district_codes <- c(1, 3, 4, 6, 9, 10, 12, 13, 15, 18, 19, 22, 26, 27, 29, 30, 32, 33, 35, 36, 38, 39,
```

```
district <- factor(district, levels = district_codes, labels = district_names)
```

```
women_edu <- factor(women_edu)
women_edu <- model.matrix(~women_edu, data = women_edu)
```

```

windex <- factor(windex)
windex <- relevel(windex, ref=3)
windex <- model.matrix(~windex, data = windex)
colnames(windex) <- c(
  "windex.Middle.ref",
  "windex.Poorest",
  "windex.Poor",
  "windex.Rich",
  "windex.Richest"
)

area <- ifelse(area==2, 0, 1)
area <- factor(area)
area <- relevel(area, ref = 1)

other_wives <- ifelse(other_wives==2, 0, 1)
other_wives <- factor(other_wives)

media <- factor(media)
media <- relevel(media, ref = 1)

```

## Hierarchical Poisson Model

### Utility functions and data

```

#####utility functions#####
## mvnrmal simulation
rmvnorm<-function(n,mu,Sigma)
{
  E<-matrix(rnorm(n*length(mu)),n,length(mu))
  t( t(E*%chol(Sigma)) +c(mu))
}

## Wishart simulation
rwish<-function(n,nu0,S0)
{
  sS0 <- chol(S0)
  S<-array( dim=c( dim(S0),n ) )
  for(i in 1:n)
  {
    Z <- matrix(rnorm(nu0 * dim(S0)[1]), nu0, dim(S0)[1]) %*% sS0
    S[, ,i]<- t(Z)%*%Z
  }
  S[, ,1:n]
}

## mvnrm log density
ldmvnorm<-function(X,mu,Sigma,iSigma=solve(Sigma),dSigma=det(Sigma))
{
  Y<-t( t(X)-mu)
  sum(diag(-.5*t(Y)%*%Y*%iSigma)) -
  .5*( prod(dim(X))*log(2*pi) + dim(X)[1]*log(dSigma) )
}

```

```

## Calculate matrix multiplication: (1,Z) %*% Beta
matmul <- function(Z, Beta){
  nZ <- nrow(Z)
  as.matrix(cbind(rep(1,nZ),Z))%*%as.matrix(Beta)
}

#####utility functions#####

#####design matrix#####
# Design matrix for all variables (random x14 | fixed x21)
X <- as.matrix(cbind(
  ### random predictors
  women_age,
  women_age2,
  husband_age,
  #husband_age2,
  HH_age,
  #HH_age2,
  women_agem,
  women_edu[,-1],
  windex[,-1],
  media,
  other_wives,
  area,
  HH_sex,
  ### fixed predictor
  division[,-1],# 7
  ethnicity
))
# Design matrix for random variabe
# (the intercept will be add later)
Z <- as.matrix(cbind(
  women_age,
  women_age2,
  husband_age,
  #husband_age2,
  HH_age,
  #HH_age2,
  women_agem,
  women_edu[,-1],
  windex[,-1],
  media,
  other_wives,
  area,
  HH_sex
))

# Concat response variable y with design matrix of X
yX <- cbind(y,X)
#####design matrix#####

```

## Init

```
#####initial values#####
q<-dim(Z)[2] + 1# number of columns in Z (13) + 1 (intercept)
p<-dim(X)[2] + 1# number of columns in X (20) + 1 (intercept)
m<-length(unique(district)) # number of district

GAMMA <- matrix(0, nrow = m, ncol = q)

# eta0: inverse-Wishart prior for Sigma
eta0 <- q+2
# The initial mean of the MVN prior of theta
model <- glm(y~., data = data.frame(yX), family=poisson)
THETA <- mu0 <- model$coefficients

# The initial variance of the MVN prior for the variance of theta
# LAMBDA is the inverse of Fisher information matrix
LAMBDA <- vcov(model)

# THETA1 stores m rows of ad hoc estimate of regression coefficients theta1
THETA1 <- NULL

# traverse each division (total of 8)
for (dis in district_codes){
  dis_data = yX[which(wm$HH7A==dis),]

  captured <- tryCatch({
    model <- glm(y~., data = data.frame(dis_data), family=poisson)
    THETA1<-rbind(THETA1, model$coeff[c(1:q)])
  }, warning = function(w) {
  })
  #print(captured)
}

S0 <- cov(THETA1)

# Prepare the list to store data in each stratum separately.
y.list <- list()
Z.list <- list()
X.list <- list()
wgt.list <- list()
N <- NULL
j <- 1
# traverse all 8 divisions
for (dis in district_codes){
  y.list[[j]] = y[which(wm$HH7A == dis)]
  N[[j]]<-length(which(wm$HH7A == dis))
  Z.list[[j]] = Z[which(wm$HH7A == dis),]
  X.list[[j]] = X[which(wm$HH7A == dis),]
  wgt.list[[j]] = wmweight[which(wm$HH7A == dis)]
  j<-j+1
}

#summary(unlist(N))
```

```

#barplot(unlist(N))

# inverse Lambda_0, inverse S_0
iL0<-iSigma<-solve(S0)

## MCMC
# The accepted count of proposed gamma_j for each stratum (j from 1 to 66)
ACCEPT.count <- rep(0,m)
# The accepted count of proposed theta
ACCEPT.Theta.count <- 0
SIGMA.post<-NULL
GAMMA.post<- list()
THETA.post <- list()
S=55000
B=5000
thin=50
#####initial values#####

#####simulation#####
set.seed(123)
start.time <- Sys.time()
for(s in 1:S)
{
  ##update Sigma
  iSigma<-rwish(1,eta0+m, solve( S0+t(GAMMA)%*%(GAMMA)) )

  ##update theta
  # theta proposed
  K_theta <- 0.05
  theta.p<-t(rmvnorm(1,THETA,K_theta*LAMBDA))
  lr<-0
  for(j in 1:m)
  {
    lr <- lr+sum(
      wgt.list[[j]] * dpois(y.list[[j]],
        lambda=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],theta.p)),log=TRUE),
      -wgt.list[[j]] * dpois(y.list[[j]],
        lambda=exp(matmul(Z.list[[j]],GAMMA[j, ]) + matmul(X.list[[j]],THETA)),log=TRUE)
    )
  }
  lr <- lr + ldmvnorm(t(theta.p),mu0,LAMBDA) - ldmvnorm(t(THETA),mu0,LAMBDA)
  #print("lr is")
  #print(lr)
  if( log(runif(1))<lr ) {
    ACCEPT.Theta.count<- ACCEPT.Theta.count+1
    THETA<-theta.p
  }

  ##update gamma
  Sigma<-solve(iSigma) ; dSigma<-det(Sigma)
  K_gamma = rep(0.1,m)
  # PD <- outer((sqrt(K_gamma)),(sqrt(K_gamma)) , "*")
  # We don't need a 66x66 matrix PD to be the scaling factor:
  # For one thing the PD %*% Sigma is non-conformable,

```

```

# and also it's enough to use a scalar as the scaling factor
# https://www.maths.lancs.ac.uk/~sherlocc/Publications/rwm.final.pdf

for(j in 1:m)
{
  # beta proposed (beta^(*)) from multivariate normal (beta_j^(s), V_J^(s))
  gamma.p<-t(rmvnorm(1,GAMMA[j,],K_gamma[j]*Sigma))
  lr<-sum(
    wgt.list[[j]] * dpois(y.list[[j]],
      lambda=exp(matmul(Z.list[[j]],gamma.p) + matmul(X.list[[j]],THETA)),log=TRUE),
    -wgt.list[[j]] * dpois(y.list[[j]],
      lambda=exp(matmul(Z.list[[j]],GAMMA[j, ] + matmul(X.list[[j]],THETA)),log=TRUE),
    ldmvnorm( t(gamma.p),rep(0,q),Sigma,iSigma=iSigma,dSigma=dSigma ),
    -ldmvnorm( t(GAMMA[j,]),rep(0,q),Sigma,iSigma=iSigma,dSigma=dSigma )
  )

  if( log(runif(1))<lr ) {
    #print(sprintf("Beta %d Accepted", j ))
    ACCEPT.count[j] <- ACCEPT.count[j]+1
    GAMMA[j,]<-gamma.p
  }
}

##store some output
if((s %% thin == 0) & (s > B)) # saving every 10th value
{
  cat("Current iteration index = ", s, "\n")
  # Print current acceptance rate
  cat("Current acceptance rate of Theta = ", round((ACCEPT.Theta.count/s)*100, digits = 2),"\n")
  # save thinned coefficients
  THETA.post<-rbind(THETA.post,t(THETA))
  SIGMA.post<-rbind(SIGMA.post,c(Sigma))
  GAMMA.post[[length(GAMMA.post)+1]]<-GAMMA
}
}

end.time <- Sys.time()
#####simulation#####

```

## Save and load results

```

RESULTPATH = "./data_output/BHP/"
save(THETA.post, file=paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
save(SIGMA.post, file=paste(RESULTPATH,"SIGMA.post.S=",S,".RData",sep=""))
save(ACCEPT.count, file=paste(RESULTPATH,"ACCEPT.count.S=",S,".RData",sep=""))
save(ACCEPT.Theta.count, file=paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))
save(GAMMA, file=paste(RESULTPATH,"GAMMA.S=",S,".RData",sep=""))
save(GAMMA.post, file=paste(RESULTPATH,"GAMMA.post.S=",S,".RData",sep=""))

RESULTPATH = "./data_output/BHP/"
load(paste(RESULTPATH,"THETA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"SIGMA.post.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.count.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"ACCEPT.Theta.count.S=",S,".RData",sep=""))

```

```
load(paste(RESULTPATH,"GAMMA.S=",S,".RData",sep=""))
load(paste(RESULTPATH,"GAMMA.post.S=",S,".RData",sep=""))
```

## Plot results

### Assess the convergence of the Markov chain

```
## stationarity plot - boxplot
stationarity.plot<-function(x,...){
  S<-length(x)
  scan<-1:S
  ng<-min( round(S/100),10)
  group<-S*ceiling( ng*scan/S) /ng
  boxplot(x~group,...)
}

# convert the result (from list) into a 47x(S-B) matrix
thin2 = 1
THETA.post.mat <- NULL
for (i in c(1:dim(THETA.post)[1])){
  if(i %% thin2 == 0){
    # Thin again
    THETA.post.mat <- cbind(THETA.post.mat, as.matrix(unlist(t(THETA.post)[,i])) )
  }
}
GAMMA.post.mat <- list()
j <- 1
for (dis in district_codes){
  gamma_j <- NULL
  for (s in c(1:length(GAMMA.post))){
    if(i %% thin2 == 0){
      # Thin again
      gamma_j <- rbind(gamma_j, GAMMA.post[[s]][j, ])
    }
  }
  GAMMA.post.mat[[j]] <- gamma_j
  j<-j+1
}
# GAMMA.post.mat Usage: matmul(Z, apply(GAMMA.post.mat[[9]],2,mean))
SIGMA.post.mat <- NULL
for (i in c(1:dim(SIGMA.post)[1])){
  if(i %% thin2 == 0){
    # Thin again
    SIGMA.post.mat <- cbind(SIGMA.post.mat, SIGMA.post[i,] )
  }
}

ACCEPT.count/S
```

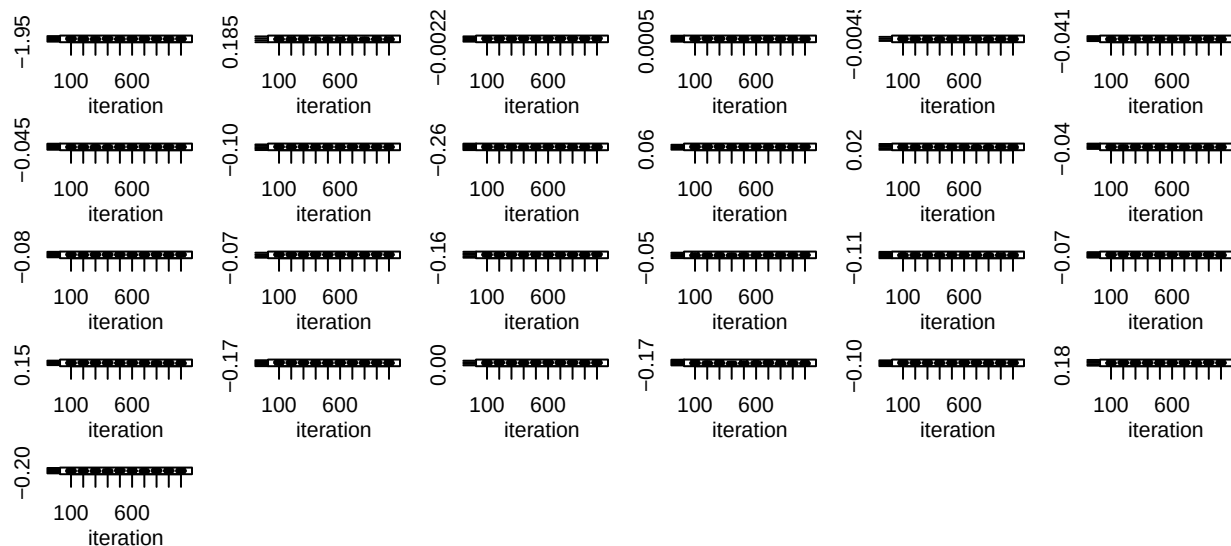
```
## [1] 0.4313273 0.4673818 0.4650727 0.3954545 0.4144727 0.3586545 0.3616909
## [8] 0.4088909 0.2782545 0.4446545 0.2971273 0.3895818 0.2640364 0.3694364
## [15] 0.4084909 0.4424727 0.3816364 0.3904182 0.4574545 0.4017091 0.4676545
## [22] 0.3995091 0.3851455 0.4806909 0.4275455 0.4687273 0.4013091 0.3696727
```

```
## [29] 0.4015818 0.4096182 0.4205818 0.4359091 0.4504364 0.4560727 0.4305091
## [36] 0.4744909 0.4105455 0.4461091 0.3137455 0.4041636 0.4692545 0.3882364
## [43] 0.3919091 0.4333091 0.4202182 0.3922909 0.4105636 0.3628182 0.3875273
## [50] 0.4587636 0.4230182 0.4515636 0.3920000 0.4447636 0.4748727 0.3780727
## [57] 0.4440545 0.4099636 0.3568545 0.4415818 0.3500727 0.3573636 0.3674545
## [64] 0.4277091
```

```
#stationarity.plot(THETA.post[, 1],xlab="iteration",ylab=expression(theta[1]))
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
for (i in 1:p) {
stationarity.plot(THETA.post.mat[i, ],xlab="iteration",ylab=expression(theta))
}
dev.print(pdf,
paste(IMAGEOUT, "THETA.post.boxplot.pdf", sep=""),
width=16,height=12)
```

```
## pdf
## 2
```

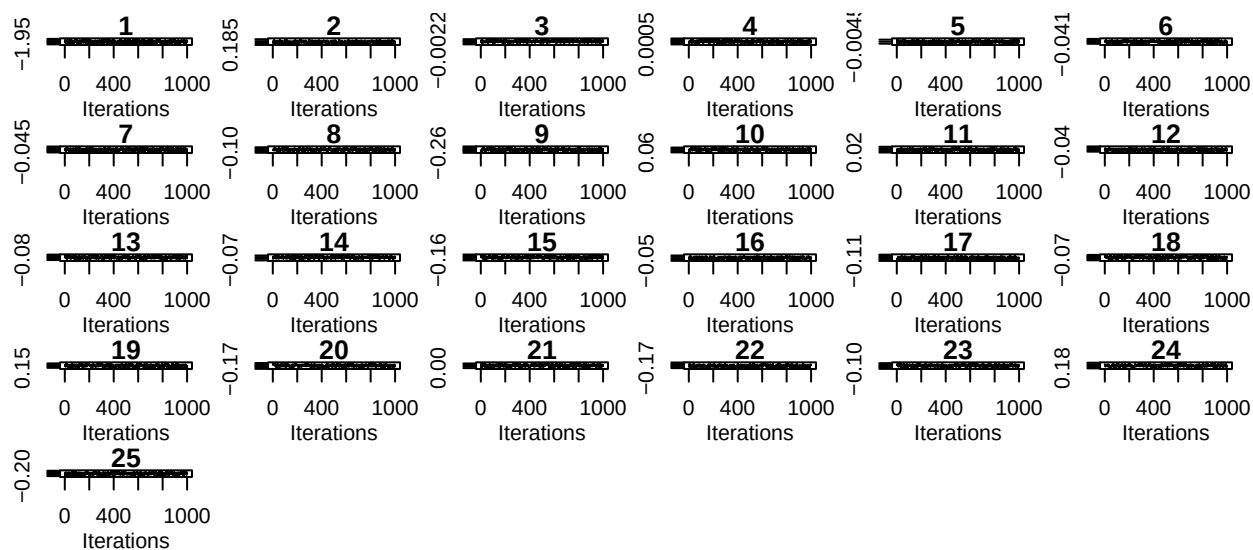
```
# Traceplot
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
```



```
for (i in 1:p) {
plot(seq(1,dim(THETA.post.mat)[2],by=1), THETA.post.mat[i,], type = "l", xlab = "Iterations", ylab = ex
}
dev.print(pdf,
paste(IMAGEOUT, "THETA.post.pdf", sep=""),
width=16,height=12)
```

```
## pdf
## 2
```

```
# ACF plot
library(coda)
par(mfrow = c(8, 6), mar=c(3,2,1,1),mgp=c(2,1,0))
```



```
for (i in 1:p){
  esize <- effectiveSize(THETA.post.mat[i,])
  acf_plot <- acf(THETA.post.mat[i,], plot = FALSE)
  t <- paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=t)
}
dev.print(pdf,
  paste(IMAGEOUT,"ACF.THETA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
## 2
```

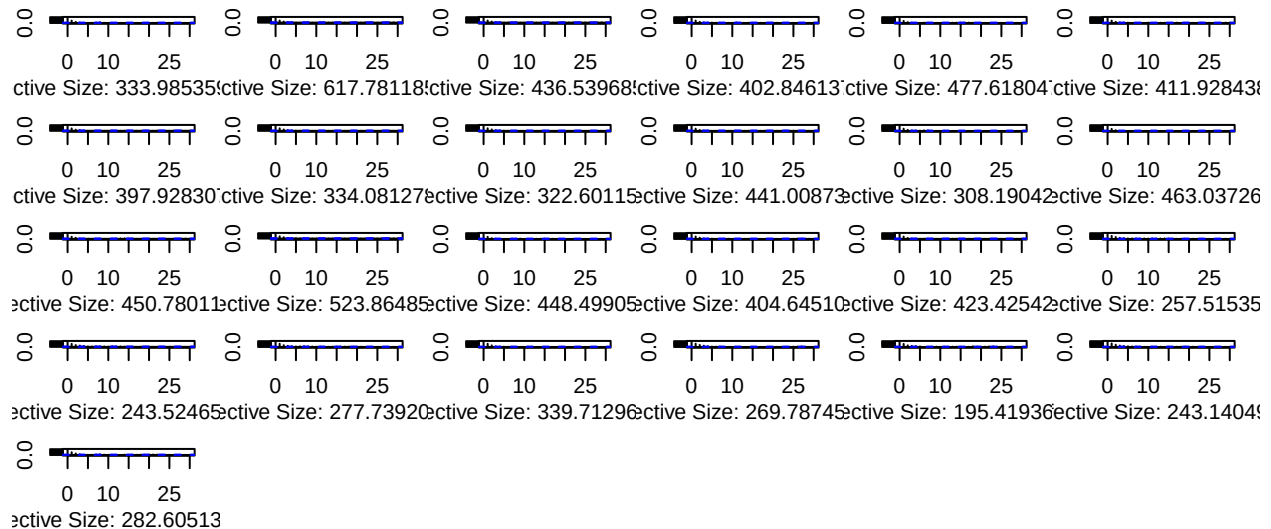
```
apply(SIGMA.post, 2, effectiveSize)
```

```
##      [1] 210.89416 164.75685 165.20749 120.08147 121.41495 150.43013 148.61536
##      [8] 168.53194 183.35733 259.56514 219.74080 149.95814 88.18227 203.75464
##     [15] 136.78390 170.92979 126.74542 164.75685 129.70084 128.64027 63.88940
##     [22] 88.26333 146.04979 71.47565 120.82544 134.15036 217.90987 115.53872
##     [29] 134.98084 89.90937 162.69113 109.43546 99.72703 100.02275 165.20749
##     [36] 128.64027 0.00000 116.43906 103.10781 153.83420 92.53562 130.58133
##     [43] 132.28458 224.82281 126.25194 149.30063 112.24977 168.57516 122.67155
##     [50] 115.10453 107.22007 120.08147 63.88940 116.43906 111.67526 148.37317
##     [57] 121.11863 140.03720 143.40157 140.63004 181.97514 136.69787 155.91693
##     [64] 157.89153 98.19185 170.45051 70.22424 86.51217 121.41495 88.26333
##     [71] 103.10781 148.37317 161.33007 155.13391 128.50098 179.28451 150.01133
##     [78] 222.03047 138.86202 152.26334 154.59124 151.32805 136.34710 130.85301
##     [85] 173.75466 150.43013 146.04979 153.83420 121.11863 155.13391 179.23519
##     [92] 112.69216 151.08016 144.99942 351.59061 166.60626 171.60288 164.68604
##     [99] 159.86064 233.36897 156.57209 123.42930 148.61536 71.47565 92.53562
##    [106] 140.03720 128.50098 112.69216 137.04634 118.46926 94.25336 237.72140
##    [113] 151.03263 167.73950 112.01218 169.10543 82.63688 88.70104 159.98707
##    [120] 168.53194 120.82544 130.58133 143.40157 179.28451 151.08016 118.46926
##    [127] 125.03800 108.88533 213.61106 161.99050 158.23211 148.74608 154.12948
##    [134] 176.37476 82.28565 141.52025 183.35733 134.15036 132.28458 140.63004
##    [141] 150.01133 144.99942 94.25336 108.88533 120.11858 209.61076 173.80940
##    [148] 176.27154 131.89565 149.60330 127.17533 126.78831 161.54704 259.56514
```

```
## [155] 217.90987 224.82281 181.97514 222.03047 351.59061 237.72140 213.61106
## [162] 209.61076 327.65014 160.84837 206.67038 200.64421 195.87846 176.84319
## [169] 209.78093 314.01910 219.74080 115.53872 126.25194 136.69787 138.86202
## [176] 166.60626 151.03263 161.99050 173.80940 160.84837 140.53642 148.22413
## [183] 121.67624 136.02660 123.05635 158.16661 181.76226 149.95814 134.98084
## [190] 149.30063 155.91693 152.26334 171.60288 167.73950 158.23211 176.27154
## [197] 206.67038 148.22413 166.44996 127.97207 164.35703 165.70929 105.94247
## [204] 191.17037 88.18227 89.90937 112.24977 157.89153 154.59124 164.68604
## [211] 112.01218 148.74608 131.89565 200.64421 121.67624 127.97207 141.41432
## [218] 154.36122 186.84909 91.68523 142.75845 203.75464 162.69113 168.57516
## [225] 98.19185 151.32805 159.86064 169.10543 154.12948 149.60330 195.87846
## [232] 136.02660 164.35703 154.36122 135.41386 157.82432 139.79193 166.54802
## [239] 136.78390 109.43546 122.67155 170.45051 136.34710 233.36897 82.63688
## [246] 176.37476 127.17533 176.84319 123.05635 165.70929 186.84909 157.82432
## [253] 97.82636 79.45627 170.13255 170.92979 99.72703 115.10453 70.22424
## [260] 130.85301 156.57209 88.70104 82.28565 126.78831 209.78093 158.16661
## [267] 105.94247 91.68523 139.79193 79.45627 102.64626 123.74667 126.74542
## [274] 100.02275 107.22007 86.51217 173.75466 123.42930 159.98707 141.52025
## [281] 161.54704 314.01910 181.76226 191.17037 142.75845 166.54802 170.13255
## [288] 123.74667 166.44964
```

```
# index of sigma
q<-dim(Z)[2] + 1# number of columns in Z + 1 (intercept)
ind_sigma <- NULL
for (i in 1:q){
  ind_sigma <- c(ind_sigma,(i-1)*q+i)
}
```

```
par(mfrow=c(4,5))
```

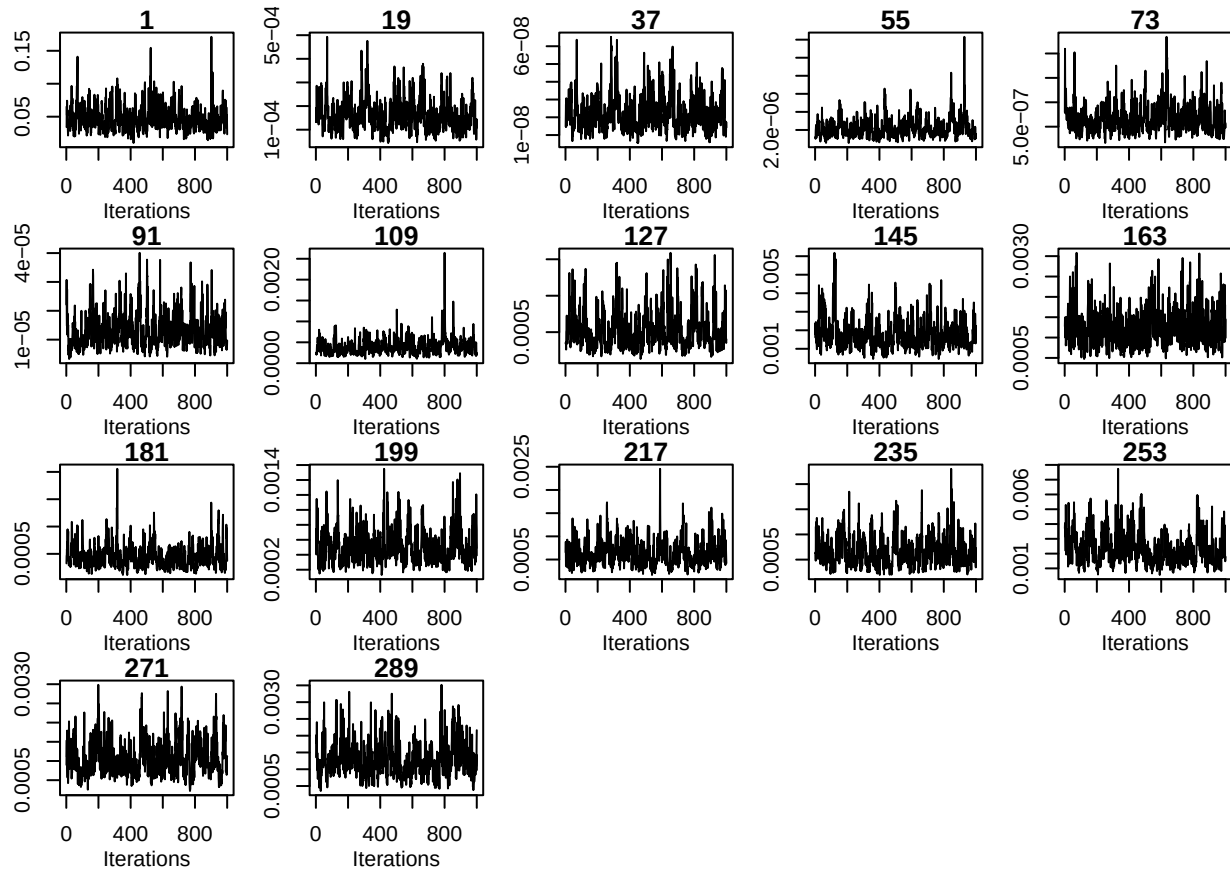


```
for(i in ind_sigma){
  plot(seq(1,dim(SIGMA.post.mat)[2],by=1), SIGMA.post.mat[i,], type = "l", xlab = "Iterations", ylab = exp)
}
dev.print(pdf,
  paste(IMAGEOUT,"SIGMA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
```

```
## 2
```

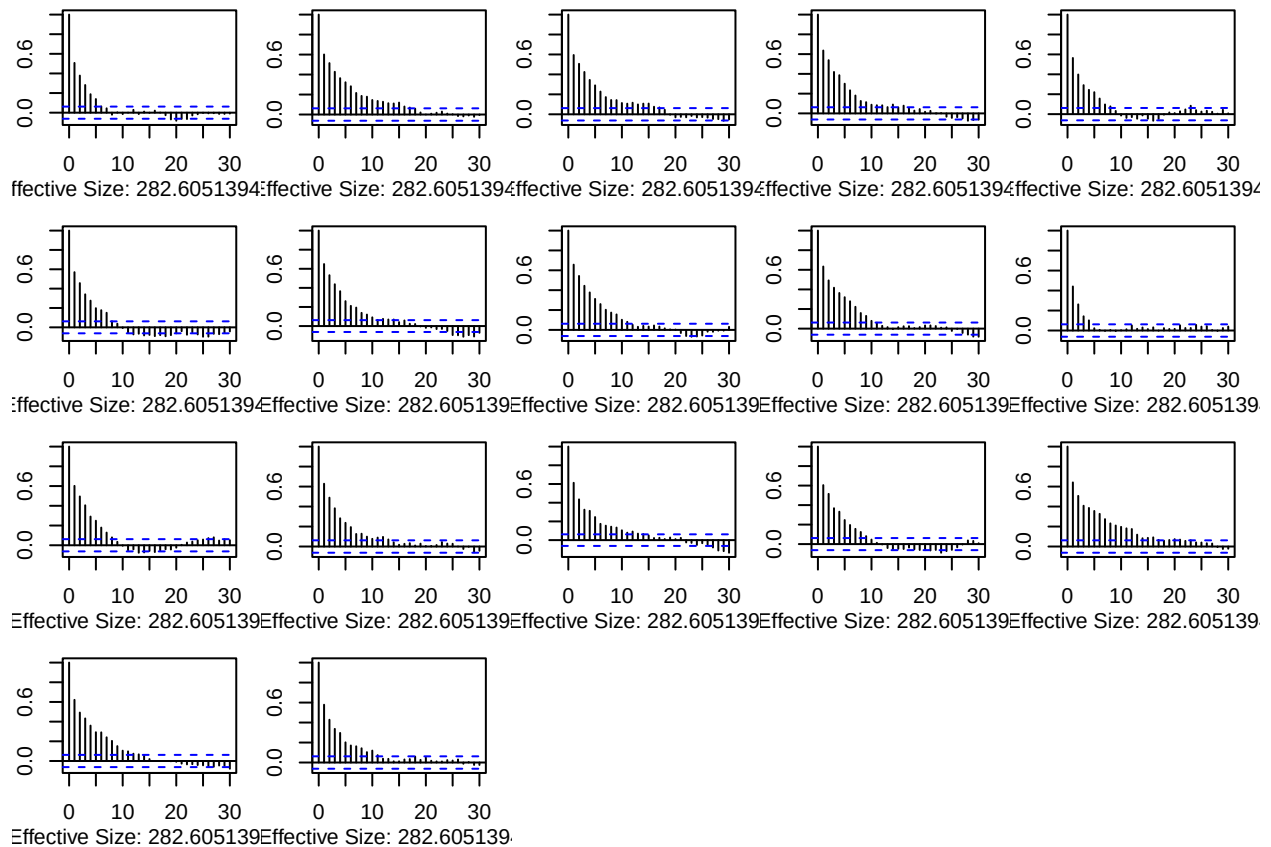
```
par(mfrow=c(4,5))
```



```
for(i in ind_sigma){
  acf_plot <- acf(SIGMA.post.mat[i,], plot=FALSE)
  t=paste(i,"Effective Size: ", esize, sep="")
  plot(acf_plot, xlab=t, ylab="", main=expression(paste("acf(",sigma^2,")",sep="")))
}
dev.print(pdf,
  paste(IMAGEOUT,"ACF.SIGMA.post.pdf",sep=""),
  width=16,height=12)
```

```
## pdf
```

```
## 2
```



### The acceptance rate of each gamma

```
library(ggplot2)
index <- 1:m
df <- data.frame(x = 1:m,
                 L = rep(0,m),
                 U = ACCEPT.count/S)
abc <- ggplot(df, aes(x = x, y = U)) +
  labs(y = expression(paste("Acceptance rate of ", gamma[j], sep="")), x = NULL) +
  geom_point(size = 2) +
  geom_linerange(aes(ymin = L)) +
  geom_hline(yintercept = 0.2, color = "red", linetype = "dashed", size = 1) +
  geom_hline(yintercept = 0.5, color = "red", linetype = "dashed", size = 1) +
  scale_y_continuous(limits = c(0, 0.8)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.

print(abc)
```



```
##      as.Date, as.Date.numeric
## Loading required package: sandwich
## Loading required package: survival
sink(paste(IMAGEOUT,"dispersion.txt",sep=""))
dispersiontest(fit.mle, alternative = "less")
sink()
```

Posterior estimates with 95% credible intervals (CI) of the incidence rate ratio (IRR) for the explanatory variables

```
library(xtable)
CI_THETA <- round(t(apply(exp(THETA.post.mat), 1, quantile ,probs=c(0.50, 0.025, 0.975))), digits = 5)
CI_THETA
```

|                   | 50%     | 2.5%    | 97.5%   |
|-------------------|---------|---------|---------|
| ## (Intercept)    | 0.16119 | 0.14649 | 0.17598 |
| ## women_age      | 1.21230 | 1.20666 | 1.21798 |
| ## women_age2     | 0.99787 | 0.99781 | 0.99794 |
| ## husband_age    | 1.00199 | 1.00105 | 1.00294 |
| ## HH_age         | 0.99597 | 0.99551 | 0.99642 |
| ## women_agem     | 0.96152 | 0.96007 | 0.96297 |
| ## women_edu1     | 0.97110 | 0.95955 | 0.98277 |
| ## women_edu2     | 0.92536 | 0.91448 | 0.93806 |
| ## women_edu3     | 0.79773 | 0.78064 | 0.81625 |
| ## windex.Poorest | 1.08649 | 1.07075 | 1.10075 |
| ## windex.Poor    | 1.03943 | 1.02595 | 1.05337 |
| ## windex.Rich    | 0.97489 | 0.96223 | 0.98854 |
| ## windex.Richest | 0.94585 | 0.93097 | 0.96179 |
| ## media          | 0.95009 | 0.94063 | 0.96082 |
| ## other_wives    | 0.88863 | 0.86738 | 0.91088 |
| ## area           | 0.97191 | 0.96085 | 0.98388 |
| ## HH_sex         | 0.92169 | 0.90581 | 0.93836 |
| ## Barishal       | 0.96396 | 0.94293 | 0.98628 |
| ## Chattogram     | 1.18447 | 1.16403 | 1.20629 |
| ## Khulna         | 0.87310 | 0.85687 | 0.88912 |
| ## Mymensingh     | 1.03472 | 1.00917 | 1.05936 |
| ## Rajshahi       | 0.86212 | 0.84535 | 0.88162 |
| ## Rangpur        | 0.93626 | 0.91922 | 0.95346 |
| ## Sylhet         | 1.24450 | 1.21598 | 1.27562 |
| ## ethnicity      | 0.85702 | 0.82796 | 0.89016 |

```
sink(paste(IMAGEOUT,"CI_THETA.tex",sep=""))
xtable(CI_THETA, digits = 3)
sink()

# BIC_h for the Hierarchical Model
q<-dim(Z)[2] + 1# number of columns in Z (13) + 1 (intercept)
p<-dim(X)[2] + 1# number of columns in X (20) + 1 (intercept)
n<-length(y)
MLE <- 0
for(j in 1:m)
{
MLE <- MLE + sum(wgt.list[[j]] * dpois(y.list[[j]], lambda=exp(matmul(Z.list[[j]], apply(GAMMA.post.mat
```

```

}
BIC_h <- q * log(m) + (p-q) * log(n) - 2 * MLE
print(BIC_h)

```

```
## [1] 148845.6
```

## Calculate the expected number of CEB

Expected Children Ever Born - Woman's age

```

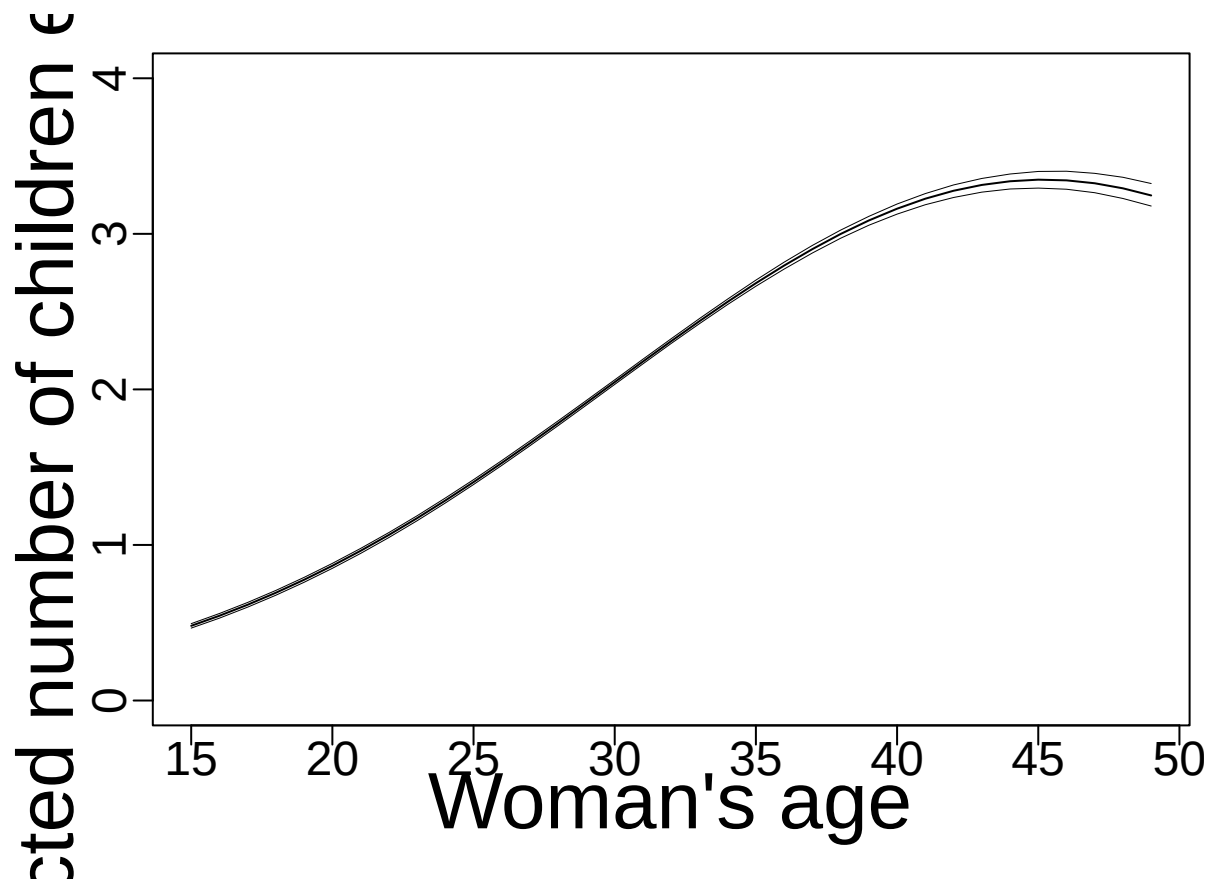
Xs <- apply(X, 2, mean)
Xs <- matrix(rep(Xs, 35), nrow = 35, byrow = T)
Xs[,1] <- seq(15, 49, by = 1)
Xs[,2] <- (seq(15, 49, by = 1)^2)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,35),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs%*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 15:49){
  sample_size <- c(sample_size, sum(X[,1]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:49)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 49),range(c(0,qE)),type="n",xlab="Woman's age",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 49, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 49, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 49, by = 1), qE[3,],col="black",lwd=0.5)

```



```
dev.print(pdf,
  paste(IMAGEOUT, "E_women_age.pdf", sep=""),
  width=16,height=16)
```

```
## pdf
## 2
```

#### Expected Children Ever Born - Husband's age

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 76), nrow = 76, byrow = T)
Xs[,3] <- seq(15, 90, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,76),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

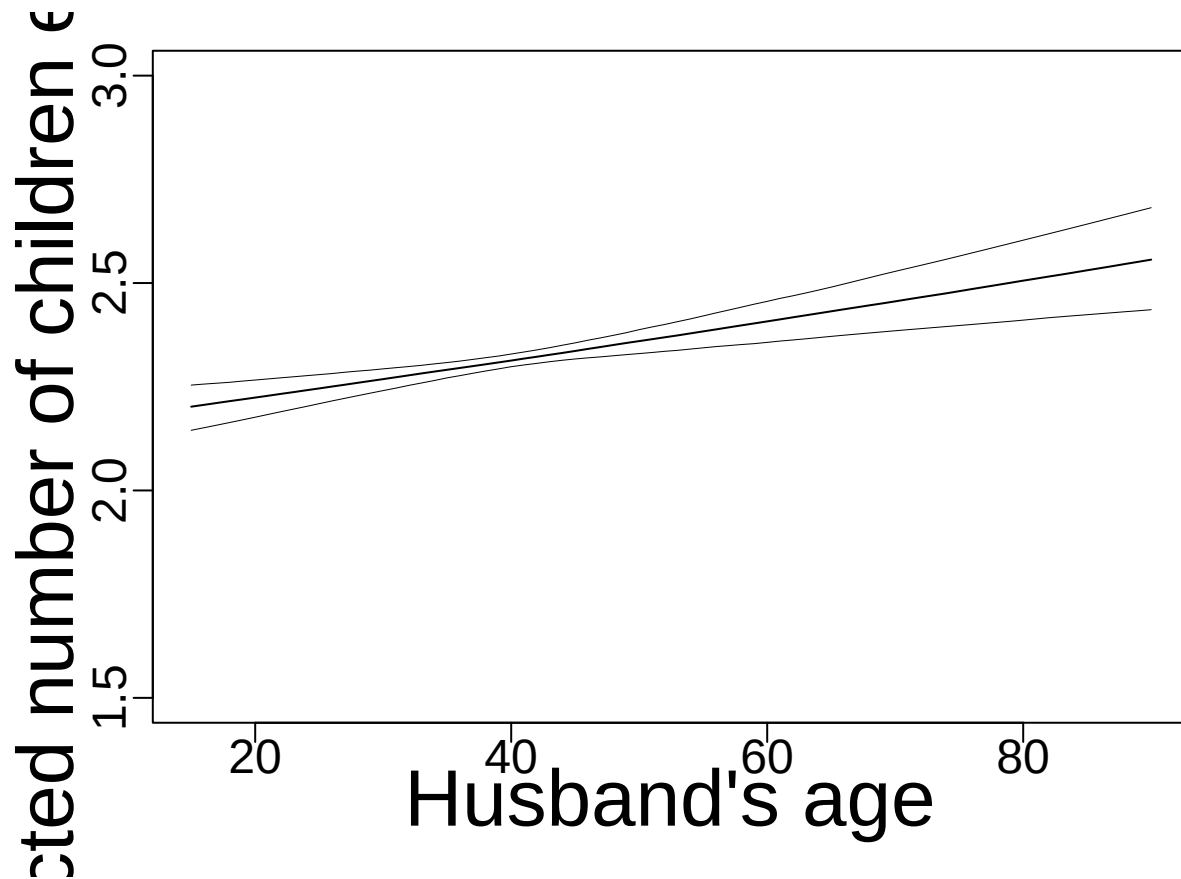
sample_size <- NULL
for (v in 15:90){
  sample_size <- c(sample_size, sum(X[,3]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(15:90)
write.csv(t(qE), paste(IMAGEOUT, "qE.husband_age.csv", sep=""))
```

```

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(15, 90),range(c(0,qE)),type="n",xlab="Husband's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(15, 90, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(15, 90, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(15, 90, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT, "E_husband_age.pdf", sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

### Expected Children Ever Born - Household head's age

```

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 80), nrow = 80, byrow = T)
Xs[,4] <- seq(16, 95, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,80),Xs) # insert the first column of 1 for intercept
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL

```

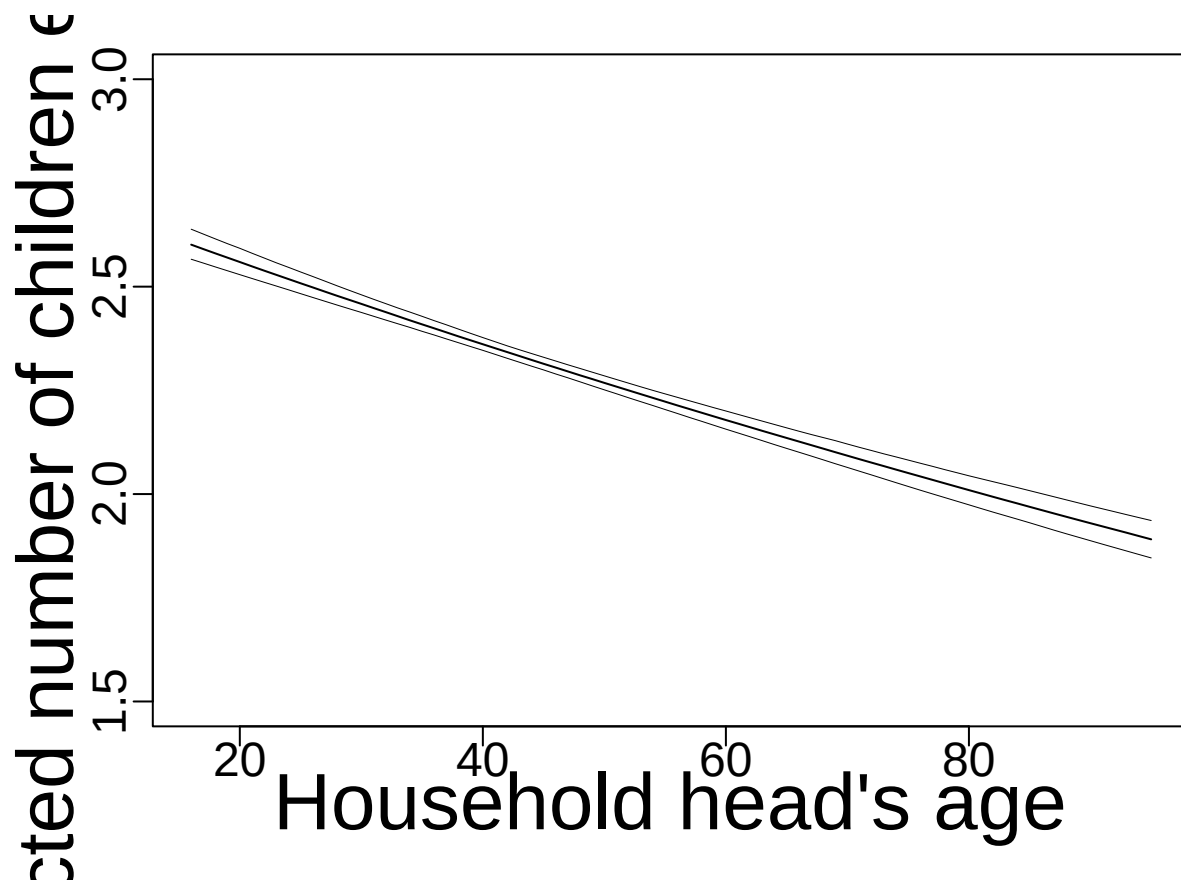
```

for (v in 16:95){
  sample_size <- c(sample_size, sum(X[,4]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(16:95)
write.csv(t(qE), paste(IMAGEOUT,"qE.HH_age.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(16, 95),range(c(0,qE)),type="n",xlab="Household head's age",ylim=c(1.5,3),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(16, 95, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(16, 95, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(16, 95, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT,"E_HH_age.pdf",sep=""),
  width=16,height=16)

```

```

## pdf
## 2

```

Expected Children Ever Born - Woman's age at marriage

```

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2

```

```

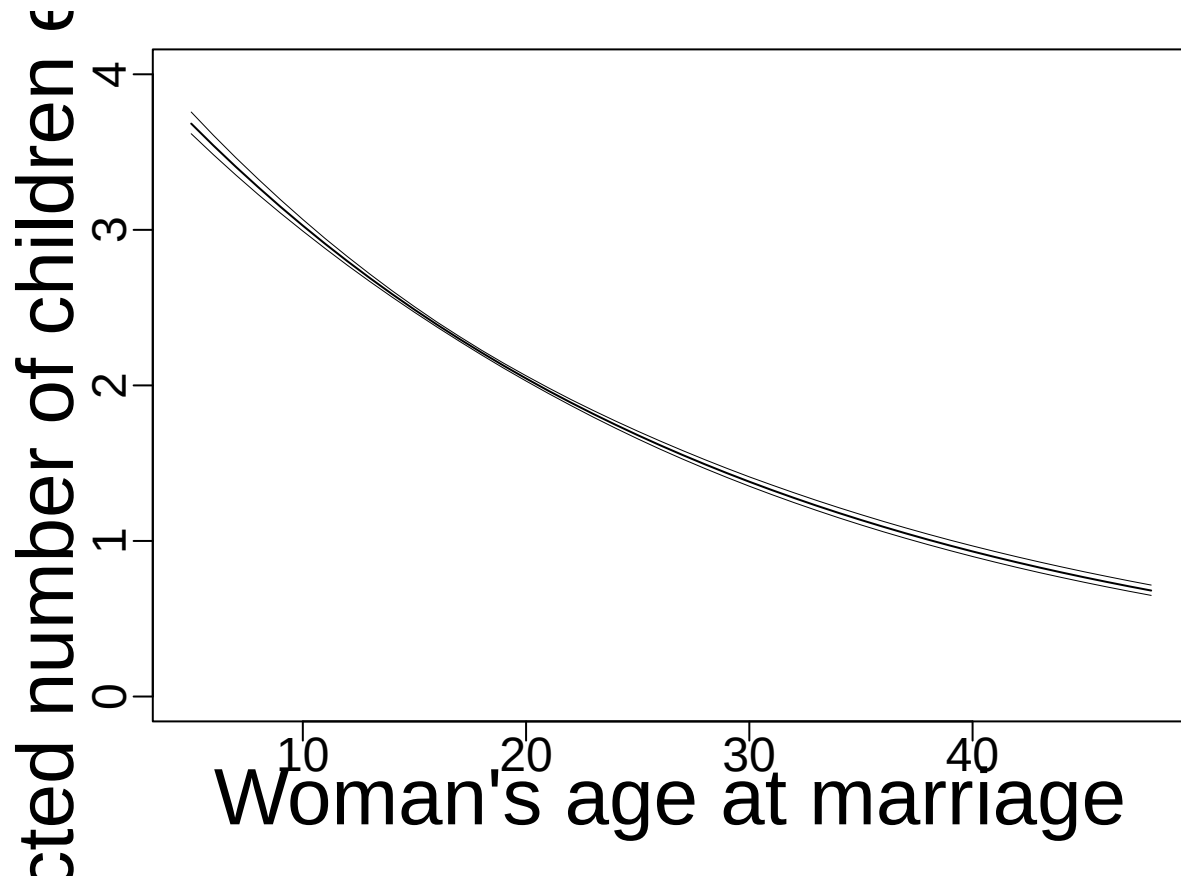
Xs <- matrix(rep(Xs, 44), nrow = 44, byrow = T)
Xs[,5] <- seq(5, 48, by = 1)
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,44),Xs)
eXB.post <- exp(Xs*%THETA.post.mat)
qE<-apply(t(eXB.post),2,quantile,probs=c(.025,.5,.975))

sample_size <- NULL
for (v in 5:48){
  sample_size <- c(sample_size, sum(X[,5]==v))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-as.character(5:48)
write.csv(t(qE), paste(IMAGEOUT,"qE.women_agem.csv",sep=""))

par(mar=c(4,4.5,1,1),mgp=c(1.88,0.50,0))
plot( c(5, 48),range(c(0,qE)),type="n",xlab="Woman's age at marriage",ylim=c(0,4),
      ylab="Expected number of children ever born", cex.lab = 2.5, cex.axis = 1.5)
lines(seq(5, 48, by = 1), qE[1,],col="black",lwd=0.5)
lines(seq(5, 48, by = 1), qE[2,],col="black",lwd=1.0)
lines(seq(5, 48, by = 1), qE[3,],col="black",lwd=0.5)

```



```

dev.print(pdf,
  paste(IMAGEOUT,"E_women_agem.pdf",sep=""),
  width=16,height=16)

```

```
## pdf
## 2
```

## Expected Children Ever Born - Division

```
# Expected Number of Children Ever Born in Bangladesh
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
eXB.post<- exp(t(t(Xs)%*%THETA.post.mat) )
qEA<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
round(qEA, digits = 3)

##          [,1]
## 2.5%    2.298
## 50%     2.313
## 97.5%   2.329

qEA <- rbind(qEA,dim(X)[1])
# Division

dnames <- c(
  "Bangladesh",
  "Dhaka",
  "Barishal",
  "Chattogram",
  "Khulna",
  "Mymensingh",
  "Rajshahi",
  "Rangpur",
  "Sylhet"
)

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- matrix(rep(Xs, 8), nrow = 8, byrow = T)
for (div_index in c(2:8)){
  x <- rep(0,8)
  x[div_index] <- 1
  Xs[, (15+div_index)] <- x
}
colnames(Xs) <- colnames(X)
Xs = cbind(rep(1,8),Xs)

eXB.post<- exp(t(Xs)%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))

sample_size <- sum(wm$HH7==30) # count number of women in Herat(reference levels)
for (HH7 in c(10,20,40,45,50,55,60)){
  sample_size <- c(sample_size, sum(wm$HH7==HH7))
}
qE <- rbind(qE,sample_size)

qE.out <- cbind(qEA,qE)
```

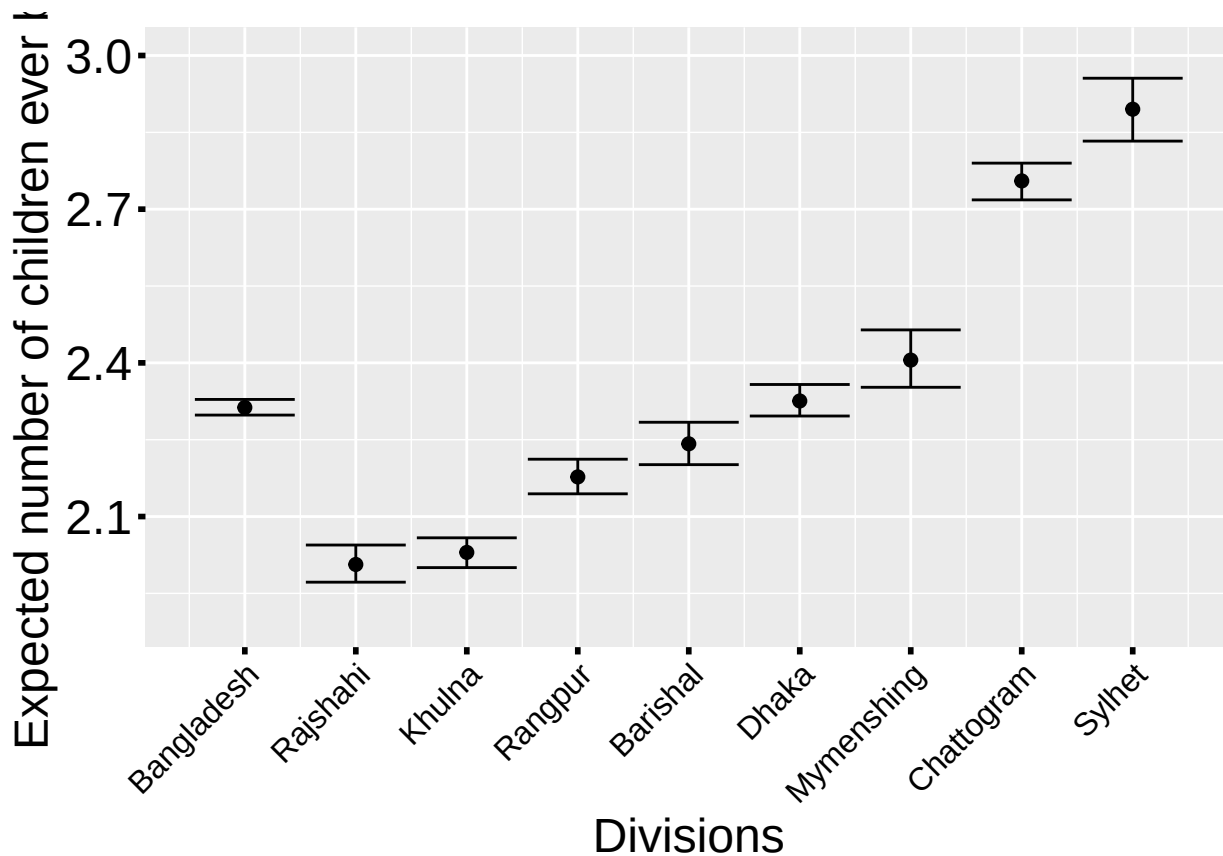
```

colnames(qE.out)<-dnames
write.csv(t(qE.out), paste(IMAGEOUT,"qE.province.csv",sep=""))

index <- order(qE[2,])
index <- c(1, index + 1)

library(ggplot2)
BHP.df_division <- data.frame(x = 1:9,
                              F = qE.out[2,][index],
                              L = qE.out[1,][index],
                              U = qE.out[3,][index],
                              name = dnames[index])
abc <- ggplot(BHP.df_division, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Divisions") +
  ylim(1.9,3.0) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:9), labels = dnames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHP.df_division, file=paste(IMAGEOUT,"BHP.df_division.RData",sep=""))
#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps",sep=""),width=8,height=8, horizontal = FALSE)
dev.print(pdf, paste(IMAGEOUT, "E_division.pdf",sep=""),width=8,height=8)

```

```
## pdf
## 2
```

### Expected Children Ever Born - Ethnicity

```
anames<- c("Bengali", "Other")
index <- c(1, 2)
anames[index]
```

```
## [1] "Bengali" "Other"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,25] <- c(1, 2) # 2 for Other, 1 for Bengali
```

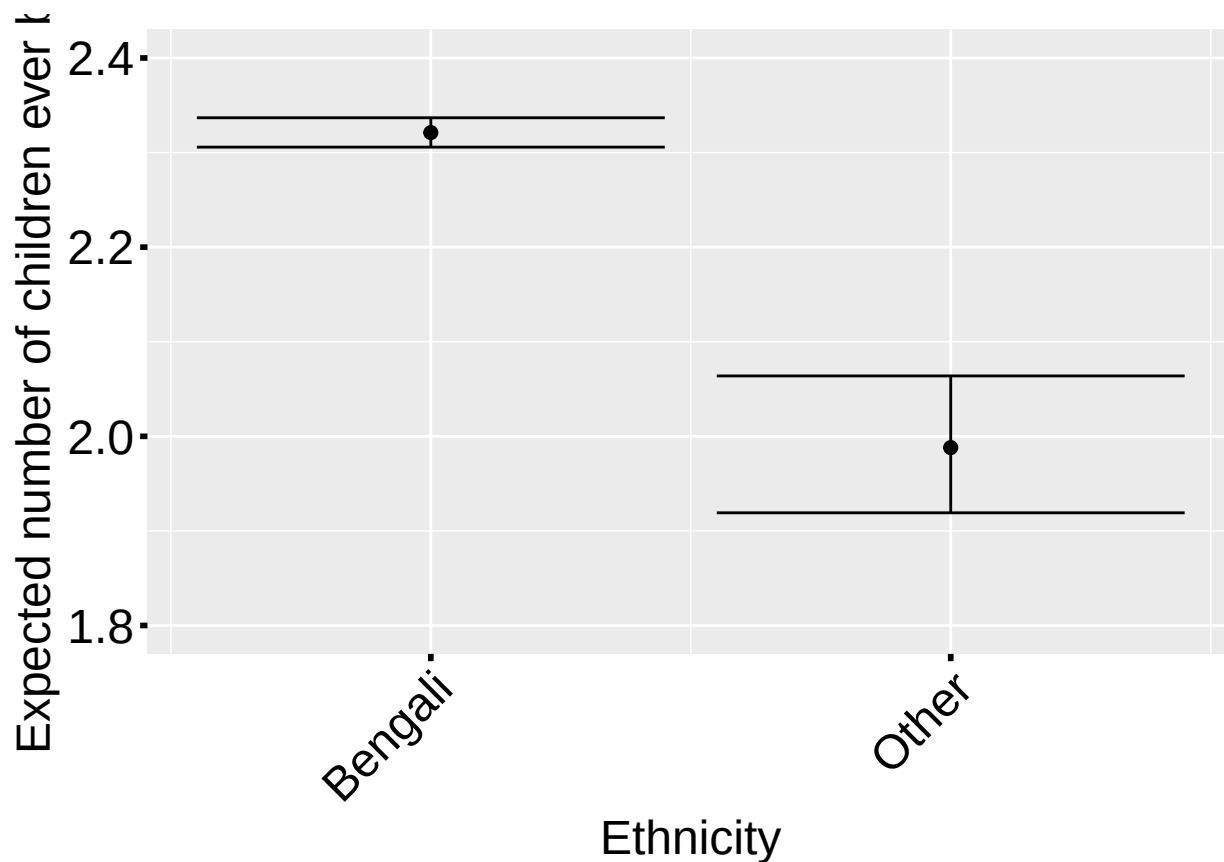
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,24]==1), sum(X[,24]==2)) # Bengali , Other
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.ethnicity.csv",sep=""))
```

```
BHP.df_ethnicity <- data.frame(x =1:2,
                              F = c(qE[2,])[index],
                              L = c(qE[1,])[index],
                              U = c(qE[3,])[index])
```

```
abc <- ggplot(BHP.df_ethnicity, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Ethnicity") +
  geom_point(size = 2) +
  ylim(1.8,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BHP.df_ethnicity, file=paste(IMAGEOUT,"BHP.df_ethnicity.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_ethnicity.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## Bengali
## 0.1675026
```

### Expected Children Ever Born - Area

```
anames<- c("Rural", "Urban")
index <- c(1, 2)
anames[index]
```

```
## [1] "Rural" "Urban"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,16] <- c(1, 2) # 2 for Urban, 1 for Rural
```

```
eXB.post<- exp(t(Xs*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

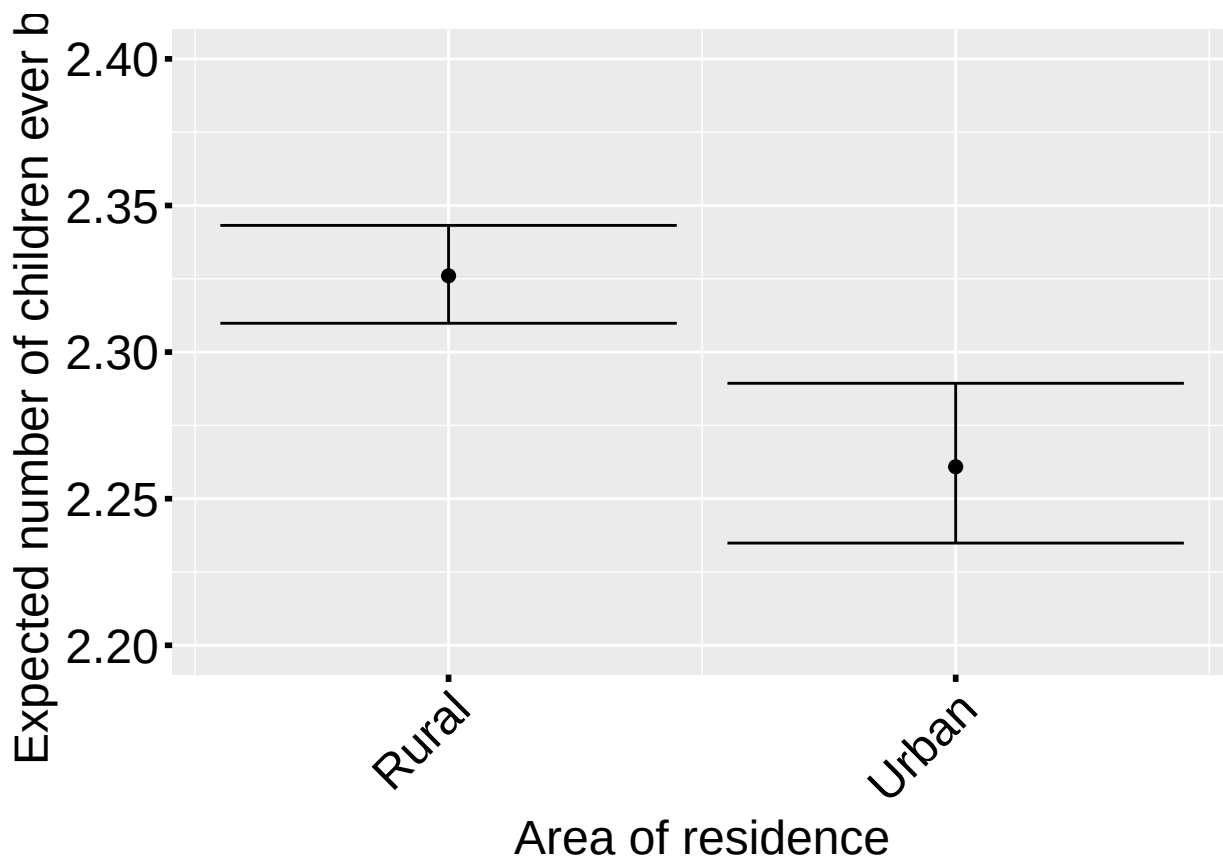
```

sample_size <- c(sum(X[,15]==1), sum(X[,15]==2)) # Rural, Urban
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.area.csv",sep=""))

BHP.df_area <- data.frame(x = 1:2,
                          F = c(qE[2,])[index],
                          L = c(qE[1,])[index],
                          U = c(qE[3,])[index])
abc <- ggplot(BHP.df_area, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Area of residence") +
  geom_point(size = 2) +
  ylim(2.2,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHP.df_area, file=paste(IMAGEOUT,"BHP.df_area.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_area.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##      Rural
## 0.02880188
```

## Expected Children Ever Born - Women's education

```
winames <- c("Pre-primary or none", "Primary", "Secondary", "Higher secondary+")
index <- c(1, 2, 3, 4)
winames[index]
```

```
## [1] "Pre-primary or none" "Primary"          "Secondary"
## [4] "Higher secondary+"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 4), nrow = 4, byrow = T)
Xs[,7] <- c(0, 1, 0, 0)
Xs[,8] <- c(0, 0, 1, 0)
Xs[,9] <- c(0, 0, 0, 1)
```

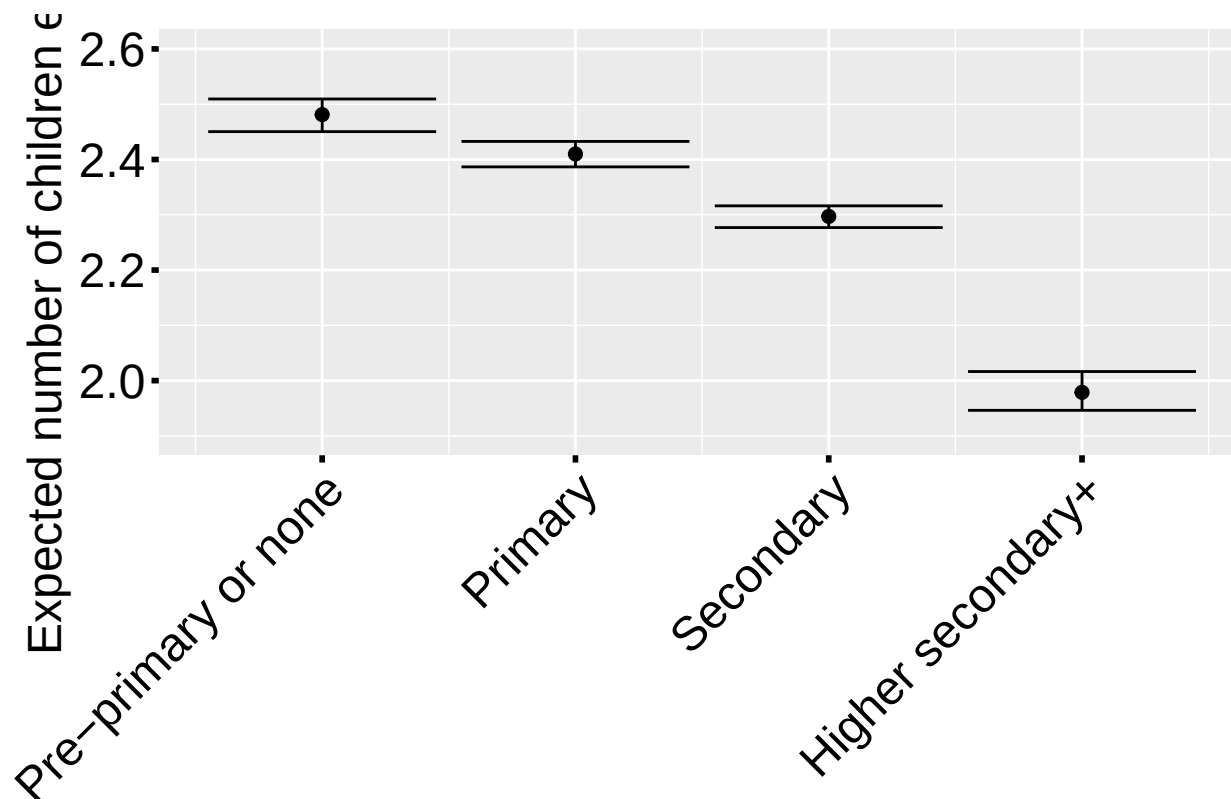
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- NULL
for (edu in 0:3){
  sample_size <- c(sample_size, sum(wm$welevel==edu))
}
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.women_edu.csv",sep=""))
```

```
BHP.df_women_edu <- data.frame(x =1:4,
                                F = c(qE[2,][index],
                                L = c(qE[1,][index],
                                U = c(qE[3,][index])
```

```
abc <- ggplot(BHP.df_women_edu, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Women's education", cex.lab = 2.5, cex.axis = 1.5) +
  geom_point(size = 2) +
  ylim(1.9,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:4), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



## Women's education

```
save(BHP.df_women_edu, file=paste(IMAGEOUT, "BHP.df_women_edu.RData", sep=""))

dev.print(pdf, paste(IMAGEOUT, "E_women_edu.pdf", sep=""), width=8, height=8)

## pdf
## 2
```

## Expected Children Ever Born - Wealth Index

```
winames <- c("Middle", "Poorest", "Poor", "Rich", "Richest")
index <- c(2, 3, 1, 4, 5)
winames[index]

## [1] "Poorest" "Poor"      "Middle"  "Rich"    "Richest"

Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 5), nrow = 5, byrow = T)
Xs[,10] <- c(0, 1, 0, 0, 0)
Xs[,11] <- c(0, 0, 1, 0, 0)
Xs[,12] <- c(0, 0, 0, 1, 0)
Xs[,13] <- c(0, 0, 0, 0, 1)

eXB.post <- exp(t(Xs*%THETA.post.mat) )
qE <- apply( eXB.post, 2, quantile, probs=c(.025, .5, .975))

sample_size <- sum(wm$windex5==3)
```

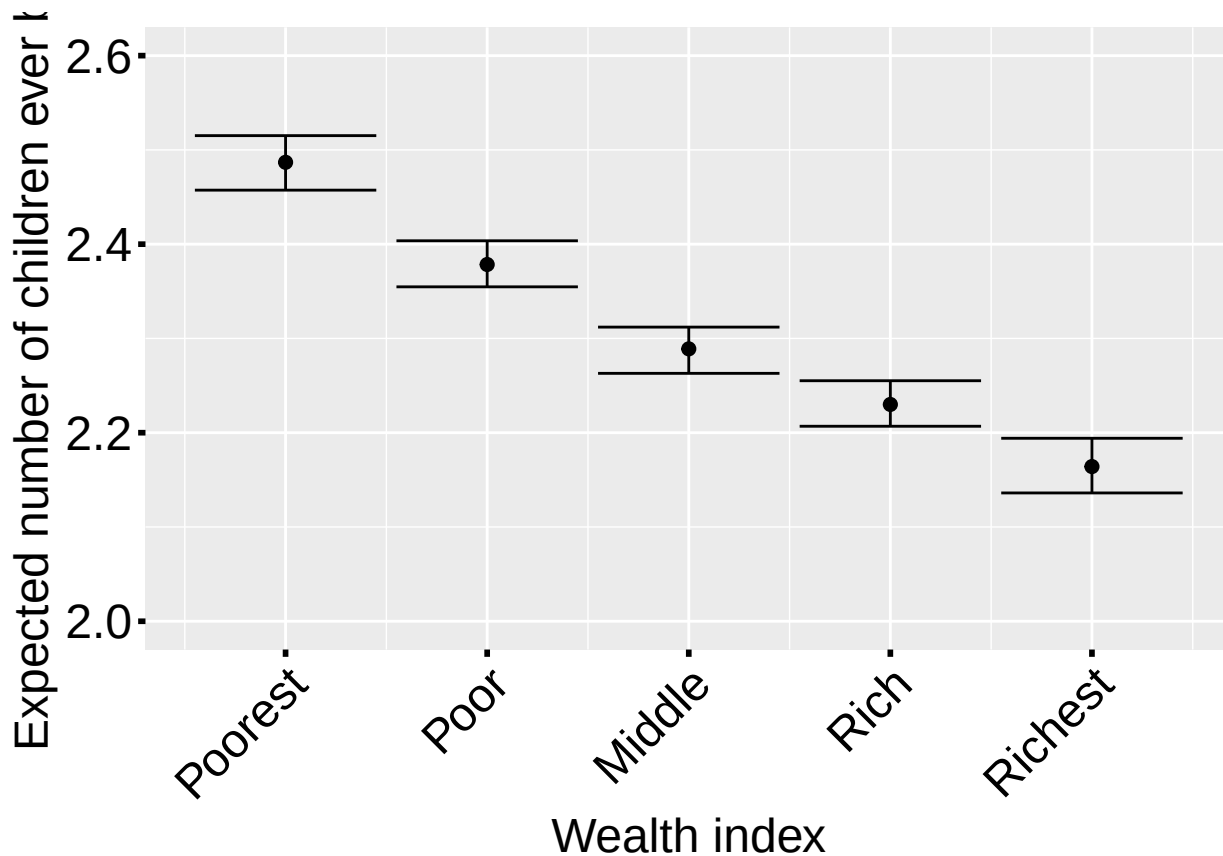
```

for (wealth in c(1,2,4,5)){
  sample_size <- c(sample_size, sum(wm$windex5==wealth))
}
qE <- rbind(qE,sample_size)

colnames(qE)<-winames
write.csv(t(qE), paste(IMAGEOUT,"qE.windex.csv",sep=""))

BHP.df_windex5 <- data.frame(x =1:5,
                             F = c(qE[2,])[index],
                             L = c(qE[1,])[index],
                             U = c(qE[3,])[index])
abc <- ggplot(BHP.df_windex5, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Wealth index", cex.lab = 2.5, cex.axis = 1.5) +
  geom_point(size = 2) +
  ylim(2,2.6) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:5), labels = winames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHP.df_windex5, file=paste(IMAGEOUT,"BHP.df_windex5.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_windex5.pdf",sep=""),width=8,height=8)

```

```
## pdf
## 2
```

### Expected Children Ever Born - other wives

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,15] <- c(1, 2) # 2 for Yes, 1 for No
```

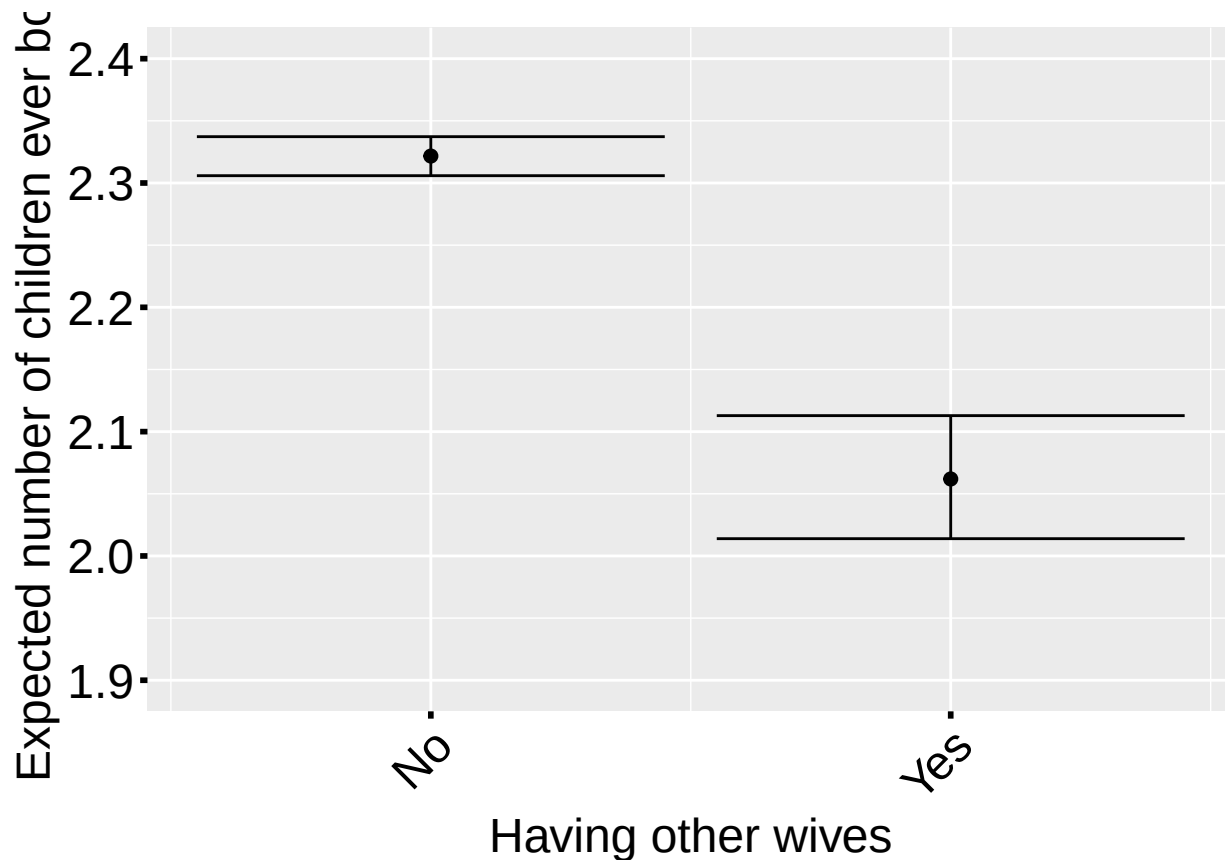
```
eXB.post<- exp(t(Xs%%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,14]==1), sum(X[,14]==2)) # No, Yes
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.other_wives.csv",sep=""))
```

```
BHP.df_other_wives <- data.frame(x =1:2,
                                F = c(qE[2,])[index],
                                L = c(qE[1,])[index],
                                U = c(qE[3,])[index])
```

```
abc <- ggplot(BHP.df_other_wives, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Having other wives") +
  geom_point(size = 2) +
  ylim(1.9,2.4) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



```
save(BHP.df_other_wives, file=paste(IMAGEOUT,"BHP.df_other_wives.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_other_wives.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## No
## 0.1259795
```

Expected Children Ever Born - media

```
anames<- c("No", "Yes")
index <- c(1, 2)
anames[index]
```

```
## [1] "No" "Yes"
```

```
Xs <- apply(X, 2, mean)
Xs[2] <- Xs[1]^2
Xs <- as.matrix(c(1, Xs))
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)
Xs[,14] <- c(1, 2) # 2 for Yes, 1 for No
```

```
eXB.post<- exp(t(Xs*%THETA.post.mat) )
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

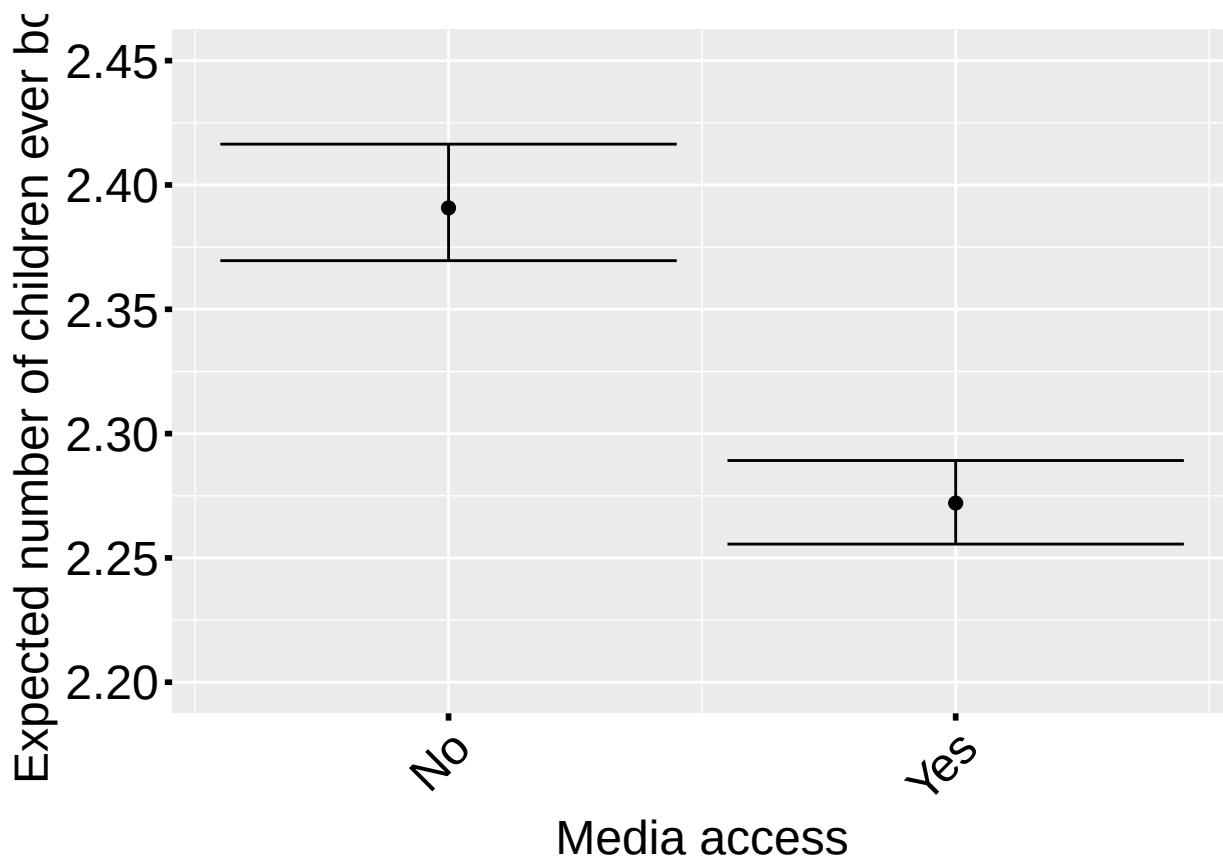
```

sample_size <- c(sum(X[,13]==1), sum(X[,13]==2)) # No, Yes
qE <- rbind(qE,sample_size)

colnames(qE)<-anames
write.csv(t(qE), paste(IMAGEOUT,"qE.media.csv",sep=""))

BHP.df_media <- data.frame(x =1:2,
                           F = c(qE[2,])[index],
                           L = c(qE[1,])[index],
                           U = c(qE[3,])[index])
abc <- ggplot(BHP.df_media, aes(x = x, y = F)) +
  labs(y = "Expected number of children ever born", x = "Media access") +
  geom_point(size = 2) +
  ylim(2.2,2.45) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

save(BHP.df_media, file=paste(IMAGEOUT,"BHP.df_media.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_media.pdf",sep=""),width=8,height=8)

```

```

## pdf
## 2

```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
##           No  
## 0.05224819
```

### Expected Children Ever Born - Household head's sex

```
anames<- c("Male", "Female")  
index <- c(1, 2)  
anames[index]
```

```
## [1] "Male"   "Female"
```

```
Xs <- apply(X, 2, mean)  
Xs[2] <- Xs[1]^2  
Xs <- as.matrix(c(1, Xs))  
Xs <- matrix(rep(Xs, 2), nrow = 2, byrow = T)  
Xs[,17] <- c(1, 2) # 2 for Female, 1 for Male
```

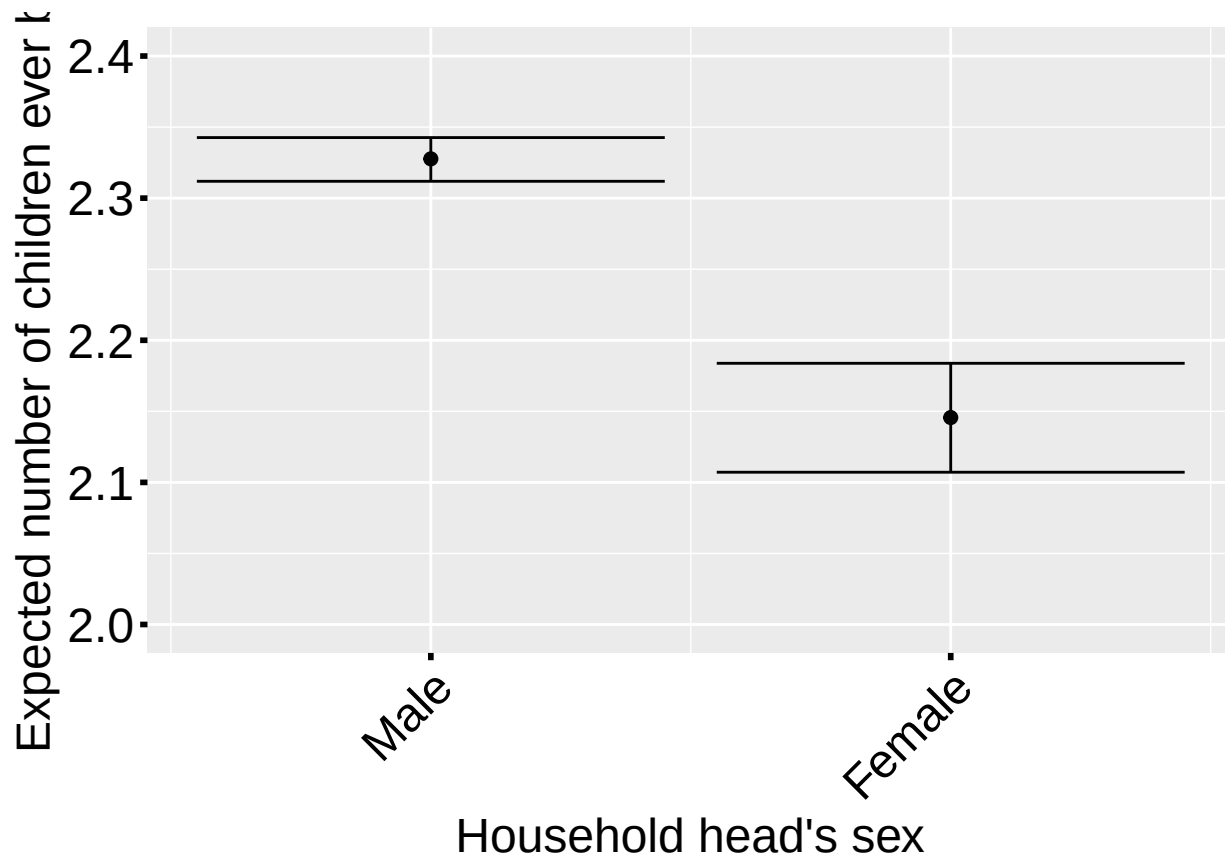
```
eXB.post<- exp(t(Xs%*%THETA.post.mat) )  
qE<-apply( eXB.post,2,quantile,probs=c(.025,.5,.975))
```

```
sample_size <- c(sum(X[,16]==1), sum(X[,16]==2)) # Male, Female  
qE <- rbind(qE,sample_size)
```

```
colnames(qE)<-anames  
write.csv(t(qE), paste(IMAGEOUT,"qE.HH_sex.csv",sep=""))
```

```
BHP.df_HH_sex <- data.frame(x =1:2,  
                             F = c(qE[2,])[index],  
                             L = c(qE[1,])[index],  
                             U = c(qE[3,])[index])
```

```
abc <- ggplot(BHP.df_HH_sex, aes(x = x, y = F)) +  
  labs(y = "Expected number of children ever born", x = "Household head's sex") +  
  geom_point(size = 2) +  
  ylim(2.0,2.4) +  
  geom_errorbar(aes(ymax = U, ymin = L)) +  
  scale_x_continuous(breaks = c(1:2), labels = anames[index]) +  
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 18, color = "black", angle = 0),  
        axis.ticks.x = element_line(color = "black", linewidth = 1),  
        axis.text.y = element_text(size = 18, color = "black"),  
        axis.ticks.y = element_line(color = "black", linewidth = 1))  
print(abc)
```



```
save(BHP.df_HH_sex, file=paste(IMAGEOUT,"BHP.df_HH_sex.RData",sep=""))
dev.print(pdf, paste(IMAGEOUT, "E_HH_sex.pdf",sep=""),width=8,height=8)
```

```
## pdf
## 2
```

```
(qE[2,][1] - qE[2,][2])/qE[2,][2]
```

```
## Male
## 0.08483649
```

## Posterior CI of GAMMA + THETA

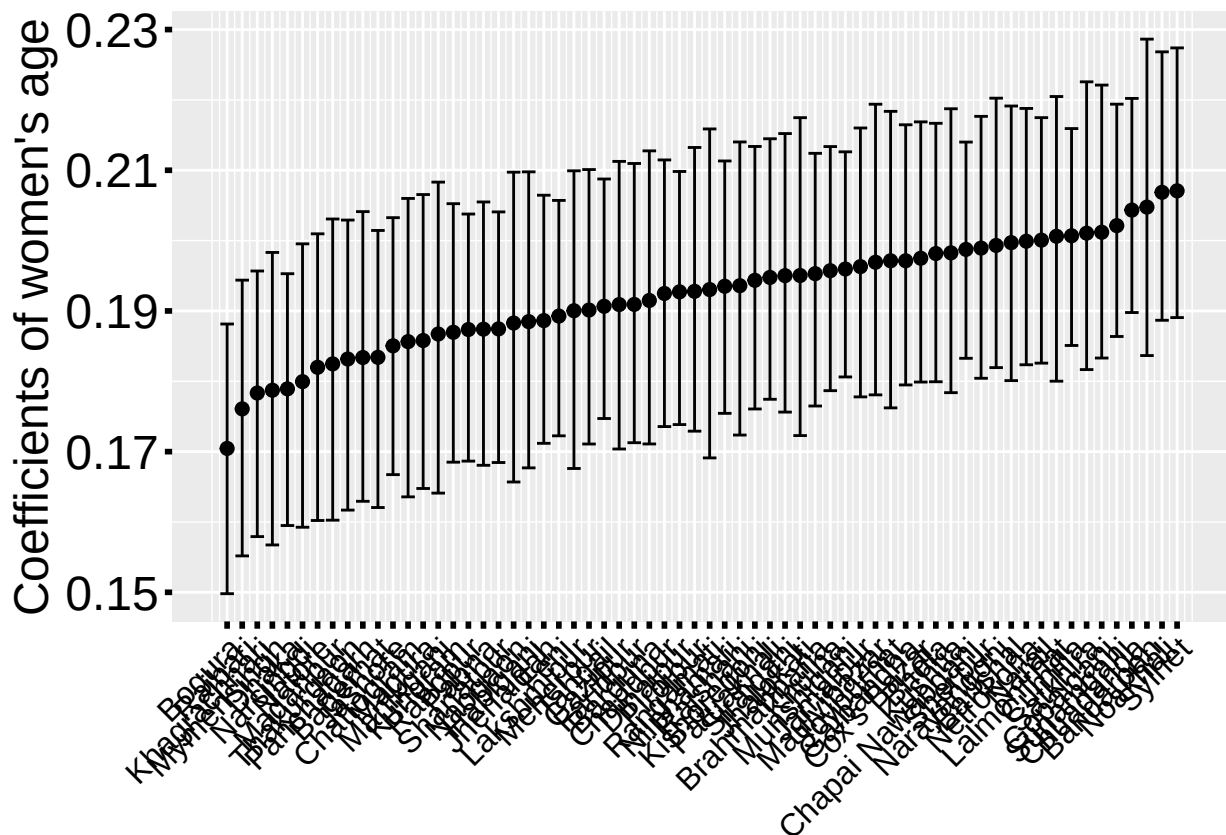
To plot the posterior CI of gamma + theta coefficients of women\_age

```
i <- 1
GAMMA_women_age <- NULL
for (dis in district_codes){
  gamma_ij <- NULL
  for (s in c(1:length(GAMMA.post))){
    gamma_ij <- c(gamma_ij, GAMMA.post[[s]][i, 2])
  }
  GAMMA_women_age <- rbind(GAMMA_women_age, gamma_ij)
  i <- i+1
}
# THETA.post.mat[2,] is the fixed effect of fixed predictor
qE<-apply(GAMMA_women_age+rep(THETA.post.mat[2,],64),1,quantile,probs=c(.025,.5,.975))
index <- order(qE[2,])
```

```

df <- data.frame(x = 1:m,
  count = unlist(N),
  F = qE[2,][index],
  L = qE[1,][index],
  U = qE[3,][index])
abc <- ggplot(df, aes(x = x, y = F)) +
  labs(y = "Coefficients of women's age", x = NULL) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
  axis.ticks.x = element_line(color = "black", linewidth = 1),
  axis.text.y = element_text(size = 18, color = "black"),
  axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)

```



```

#dev.print(device = postscript, paste(IMAGEOUT, "E_province.eps", sep=""), width=8, height=8, horizontal = "t")
dev.print(pdf, paste(IMAGEOUT, "BETA.women_age.pdf", sep=""), width=16, height=8)

```

```

## pdf
## 2

```

```

write.csv(df, paste(IMAGEOUT, "qE.BETA.women_age.csv", sep=""))

```

To plot the posterior CI of gamma + theta coefficients of husband\_age

```
i <- 1
GAMMA_husband_age <- NULL
for (dis in district_codes){
  gamma_ij <- NULL
  for (s in c(1:length(GAMMA.post))){
    gamma_ij <- c(gamma_ij, GAMMA.post[[s]][i, 4])
  }
  GAMMA_husband_age <- rbind(GAMMA_husband_age, gamma_ij)
  i <- i+1
}
# THETA.post.mat[2,] is the fixed effect of fixed predictor
qE<-apply(GAMMA_husband_age+rep(THETA.post.mat[2,],64),1,quantile,probs=c(.025,.5,.975))
index <- order(qE[2,])

df <- data.frame(x = 1:m,
                 count = unlist(N),
                 F = qE[2,][index],
                 L = qE[1,][index],
                 U = qE[3,][index])
abc <- ggplot(df, aes(x = x, y = F)) +
  labs(y = "Coefficients of husband's age", x = NULL) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymax = U, ymin = L)) +
  scale_x_continuous(breaks = c(1:m), labels = district_names[index]) +
  theme(axis.title=element_text(size=18), axis.text.x = element_text(size = 12, color = "black", angle = 45),
        axis.ticks.x = element_line(color = "black", linewidth = 1),
        axis.text.y = element_text(size = 18, color = "black"),
        axis.ticks.y = element_line(color = "black", linewidth = 1))
print(abc)
```



